

Cloud Computing Note

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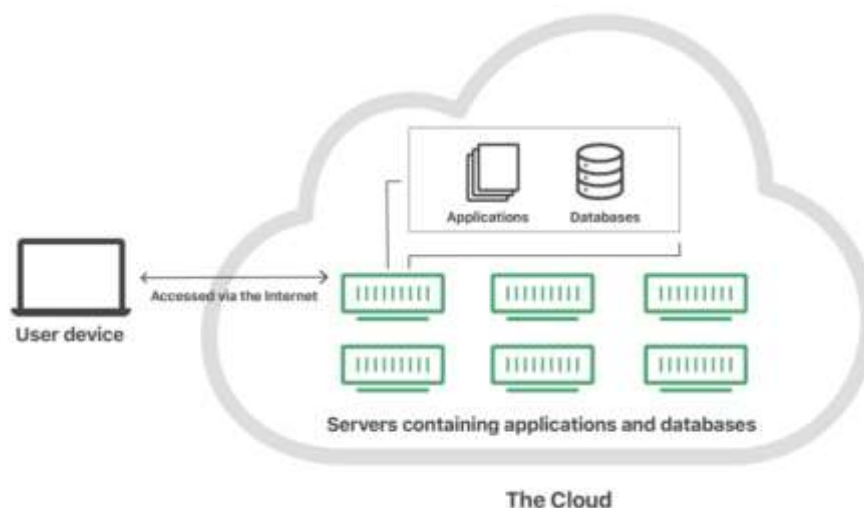
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Unit 1

Introduction

Cloud:

The term “cloud” refers to servers that are accessed over the Internet, along with the software and databases running on those servers. These cloud servers are hosted in data centers around the world, allowing users and organizations to use computing resources without managing physical servers or installing applications locally on their devices.



Note:

*In the early days of the Internet, network diagrams often represented the Internet infrastructure and servers using a cloud-shaped symbol. Because of this representation, Internet-based computing services became known as **cloud computing**.*

Cloud Computing:

The delivery of computing services such as servers, storage, databases, networking, and software through the Internet is called cloud computing. Instead of storing files and programs on a local computer on hard drive, cloud computing allows users to store and access them from remote servers through the internet. It provides flexible and on-demand access to resources without the need to own or maintain physical infrastructure.

In cloud computing, data storage and processing are done on remote servers instead of a user's local device. This makes it possible to access the same files, applications, and services from almost any device with an Internet connection.

For example, when a user logs into Instagram on a new phone after losing or changing their old device, they can still access their photos, videos, messages, and account data because everything is stored in the cloud. Similarly, services like Google Gmail, Microsoft Office 365, Dropbox, and Google Drive allow users to store and access data online anytime and from anywhere.

The advantages of Cloud computing are:

- 🚦 Cost saving as there is no need to purchase expensive hardware
- 🚦 Pay-as-you-go model, where users pay only for the resources they use
- 🚦 Easy scalability and flexibility according to user needs

[IMP one]

- ✚ Access to data and applications from anywhere with Internet access
- ✚ Faster deployment and improved efficiency
- ✚ Better backup, recovery, and security of data
- ✚ Reduced maintenance and management efforts

Characteristics of Cloud Computing:

The characteristics of cloud computing are:

- ✚ **On-demand self-service:** Users can access computing resources like storage, servers, and software whenever they need them without other human intervention.
- ✚ **Broad access:** Cloud services can be accessed through the Internet using devices like laptops, mobile phones, and tablets.
- ✚ **Resource pooling:** Computing resources are shared among multiple users and dynamically assigned according to demand.
- ✚ **Scalability and flexibility:** Resources can be easily increased or decreased based on user requirements.
- ✚ **Measured service:** Usage of cloud resources is monitored, controlled, and billed based on consumption.
- ✚ **Pay-as-you-go model:** Users only pay for the resources they actually use, reducing unnecessary costs.
- ✚ **High availability:** Cloud services are designed to remain accessible most of the time with minimal downtime.
- ✚ **Cost efficiency (economical):** Reduces the need for buying and maintaining expensive hardware and infrastructure.
- ✚ **Security:** Cloud providers use advanced security measures like encryption, backups, and authentication to protect data.
- ✚ **Easy maintenance:** Updates, maintenance, and system management are handled by the cloud provider, not the user.

Various component of Cloud computing:

[or, Participants of Cloud]

Cloud computing architecture mainly consists of three components: Clients, Data centers, and Distributed Servers. These components work together to deliver cloud services over a network

1. Clients:

Clients are the users' devices that access cloud services, such as desktops, laptops, mobile phones, and tablets. They usually connect through a Local Area Network (LAN) or the Internet. Clients are responsible for interacting with cloud applications and sending/receiving data from cloud servers.

2. Data Centers:

A data center is a large collection of powerful servers that store applications, data, and services. It provides computing power and storage for cloud users. Modern data centers use virtualization, allowing multiple virtual servers to run on a single physical server, making management efficient and scalable.

3. Distributed Servers:

Distributed servers are cloud servers located in different geographical locations. Even though they are physically far apart, they work together as a single system to provide fast and reliable services. This distribution improves performance, reduces delay, and increases availability for users worldwide.

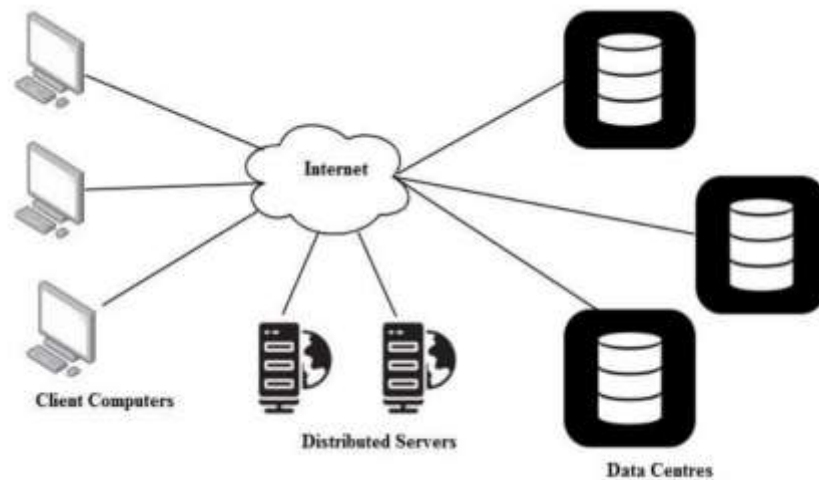
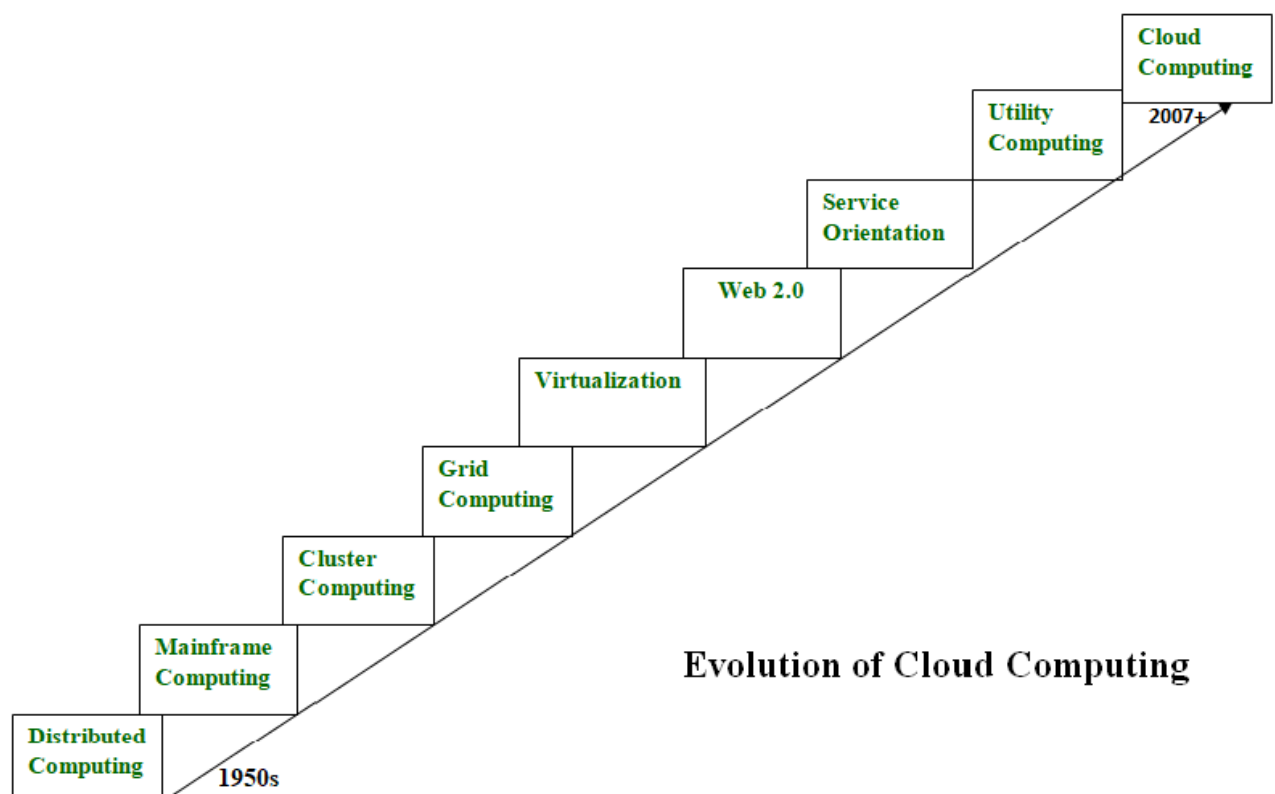


Fig: Components of Cloud computing

Evolution of Cloud Computing:

Cloud computing is based on the idea of renting computing resources over the Internet, a concept that began developing in the 1950s. Its evolution has been shaped by several important technologies, including distributed systems, mainframe computing, cluster computing, grid computing, virtualization, Web 2.0, service orientation, and utility computing.



1. Distributed System:

Distributed systems consist of multiple independent computers that work together as a single system for users. They are used to handle large-scale tasks such as massive input/output operations and online transactions. These systems offer high scalability, concurrency, and fault tolerance with very little downtime. However, they were expensive and often required systems to be located in the same geographical area. To overcome these limitations, computing evolved into mainframe, cluster, and grid systems.

2. Mainframe Computing:

Mainframe computing, introduced in 1951, consists of highly powerful and reliable computers designed for large-scale data processing. These systems provide high performance, stability, and reliability, but they are very expensive and less flexible compared to modern systems.

3. Cluster computing:

Cluster computing emerged in the 1980s as an alternative to mainframe systems. It connects multiple computers using a high-bandwidth network so that they work together as a single system. Cluster systems were much cheaper than mainframes and provided similar computing power. They also allowed easy addition of new nodes when required. However, they still required all machines to be in the same geographical location, so the location limitation was not fully solved.

4. Grid Computing:

Grid computing is a system where many computers located in different places are connected through the Internet and work together as a single system. These computers may belong to different organizations, making the system heterogeneous (different types of machines). The main idea is to share resources and solve large problems by dividing work among many systems.

It improved resource sharing across locations, but it faced challenges like network delays, low bandwidth, and communication issues. Because of these limitations, cloud computing later became more popular and is often called the successor of grid computing.

5. Virtualization:

Virtualization is a technology that allows a single physical computer to act like multiple separate computers by creating virtual machines. It adds a virtual layer between hardware and software, so multiple operating systems and applications can run at the same time on one physical system.

This improves hardware usage, reduces cost, and increases flexibility. Virtualization is the core foundation of cloud computing and is used in systems like virtual servers in cloud platforms.

6. Web 2.0:

Web 2.0 refers to the modern version of the web that allows users to interact, create, and share content. Unlike early static websites, Web 2.0 enables dynamic and interactive web applications where users can actively participate.

It supports features like social networking, online collaboration, and real-time communication. Examples include Facebook, Twitter, and Google Maps. It plays an important role in cloud computing by providing user-friendly interfaces for cloud services.

7. Service Orientation:

Service orientation is a software design approach where applications are built as a collection of services that can be reused and combined. It focuses on providing software as services instead of single large applications.

It introduces important concepts like Software as a Service (SaaS) and Quality of Service (QoS), including Service Level Agreements (SLA), which ensure performance, reliability, and availability. This model helps cloud systems stay flexible, scalable, and easy to manage.

8. Utility Computing:

Utility computing is a computing model in which computing resources such as processing power, storage, and software are provided to users as services over the Internet on a pay-per-use basis. This means users do not need to own infrastructure; instead, they rent resources whenever needed and pay only for what they actually use.

It works like public utilities such as electricity or water, where you are billed based on consumption. Utility computing helps reduce cost, improve resource utilization, and provides flexible access to computing power. It is one of the key ideas that influenced modern cloud computing pricing models like pay-as-you-go.

Thus, cloud computing is the result of continuous evolution of different computing technologies. It combines the ideas of distributed systems, virtualization, Web 2.0, service-oriented architecture, and utility computing to provide scalable, flexible and on-demand computing services over the internet. Today, cloud computing has become the backbone of modern IT services and applications used worldwide.

Challenges of Cloud Computing:

[Disadvantages/ Limitations]

The challenges of cloud computing are given as;

-  **Data confidentiality risk**
There is always a risk that user data may be accessed by unauthorized users. Therefore, strong security measures are required to protect data privacy and confidentiality.
-  **Dependence on Internet connection**
Cloud computing fully depends on the Internet. If the connection is slow or unavailable, users cannot access cloud services or data.
-  **Security concerns**
Since cloud services operate online, there is risk of cyberattacks. Issues like hacking can lead to data loss, leakage, or misuse.
-  **Compliance issues**
Cloud providers may not fully follow all legal, regulatory, or industry rules required by users or organizations.
-  **Data mobility problem**
The ability to easily access, retrieve move, and transfer data between different cloud platform or services without difficulty is called data mobility. Data mobility becomes an issue because data is not always easy to move due to differences in cloud systems, formats, and technologies. This can lead to Vendor lock-in, where users become dependent on a single provider and find it difficult to switch to another service. In addition, transferring a large amount of data takes time, cost extra money, and create compatibility issues.

Technical issues and support limitations

Users depend on cloud providers for technical support. Sometimes support may not be available 24/7 or may require additional cost.

Low connection performance

Cloud computing performance reduces when internet speed is low. Poor connectivity can lead to delays, downtime, or service disruption.

Grid Computing:

Grid computing is a type of distributed computing where a network of computers works together to complete a complex task that a single machine cannot handle efficiently. All computers in the grid follow the same communication protocol and act like a virtual supercomputer.

It is mainly used for processing large datasets, scientific calculations, and simulations that require high computing power. In this system, each computer contributes its resources such as processing power and storage. Grid computing can also be seen as a form of parallel computing where resources are spread across different locations instead of being in a single machine.

Working of Grid Computing

A grid computing system mainly consists of three types of machines:

Control Node:

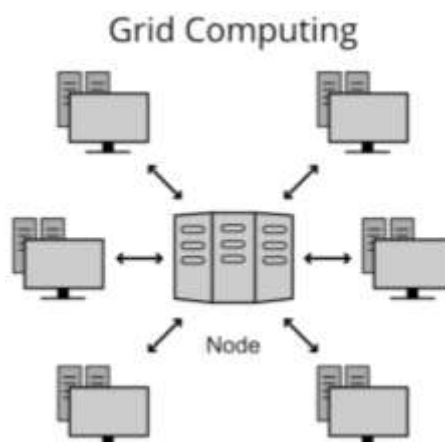
It is a server or group of servers that manages the entire grid system. It controls resources and keeps track of all activities in the network.

Provider:

These are computers that contribute their resources such as CPU power and storage to the grid. Their computers become part of the grid. All the computers shown below in the figure are providers.

User:

These are computers that use the grid resources to perform a task. They do not contribute resources.



Difference Between:

Cloud Computing	Grid Computing
➤ Uses a client–server model, where users access services from centralized cloud servers	➤ Uses a distributed computing model, where many independent systems work together
➤ Provides very high scalability, resources can increase or decrease on demand	➤ Provides limited/moderate scalability, depends on available distributed resources
➤ Offers high flexibility and on-demand services	➤ Offers less flexibility compared to cloud systems
➤ Managed through a centralized control system by cloud providers	➤ Managed in a decentralized way across multiple organizations
➤ Resources are owned by cloud service providers (Ex: AWS, Google Cloud)	➤ Resources are owned and shared by different organizations or institutions
➤ Delivers services in the form of IaaS, PaaS, SaaS	➤ Focuses on sharing computing power for large tasks and applications
➤ Mainly service-oriented (focus on delivering services to users)	➤ Mainly application-oriented (focus on solving large computational problems)
➤ Accessed using web technologies like HTTP/HTTPS and APIs	➤ Accessed using grid middleware software
➤ Resources are centrally stored and managed in data centers	➤ Resources are geographically distributed across many systems
➤ Commonly used for web hosting, storage, AI services, and applications	➤ Commonly used for scientific research, simulations, and heavy computations

Distributed Computing in Grid and Cloud:

- 🌐 Distributed computing refers to a system where multiple computers are connected through a network and work together to solve a single problem at the same time.
- 🌐 These computers form a cluster or distributed environment, often used for scientific, technical, or data-intensive tasks that require high processing power or large data handling.
- 🌐 The main goal of distributed computing is to improve performance and efficiency by sharing workloads across multiple systems in a cost-effective, transparent, and reliable way.
- 🌐 It also provides fault tolerance, meaning if one system fails, the remaining systems continue working without stopping the entire process.
- 🌐 Both cloud computing and grid computing are based on distributed computing principles, meaning they both use multiple connected systems to perform tasks.
- 🌐 In simple terms, distributed computing is the foundation for both grid computing and cloud computing.

Cloud Deployment Model:

A model that describes where the cloud is hosted, who can use it, and how it is managed is called cloud deployment model. There are 4 types of cloud deployment model. They are:

1. Private Cloud
2. Public Cloud
3. Hybrid cloud
4. Community cloud

Private Cloud:

[Internal cloud or Corporate cloud model]

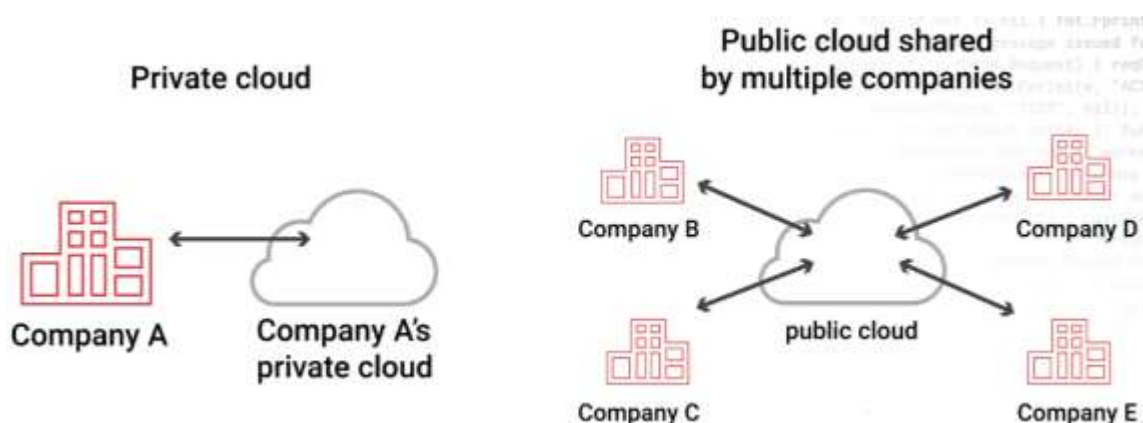
A private cloud is a cloud environment (server, data center, or network) that is dedicated to a single organization only. The cloud environment is usually owned by the organization itself. It is not shared with other organizations. It is also called the internal cloud or corporate cloud.

The advantages of Private cloud are:

- 🚦 **Customization:** Organizations can design and modify the cloud according to their specific needs and requirements.
- 🚦 **Data Privacy:** Data is accessible only to authorized internal users, making it ideal for storing sensitive corporate information.
- 🚦 **High Security:** Resources can be separated and controlled within the same infrastructure, improving access control and security levels.
- 🚦 **Full Control:** The organization has complete control over operations, policies, integrations, and user management.
- 🚦 **Supports Legacy Systems:** It can run older applications that may not work properly on public cloud platforms.

The disadvantages of Private cloud are:

- 🚦 **High Cost:** Requires large investment in hardware, software, infrastructure, and skilled staff.
- 🚦 **Limited Scalability:** Expansion depends on the available physical hardware resources, making scalability less flexible.
- 🚦 **High Maintenance:** The organization itself is responsible for maintenance, updates, and management, which increases workload and cost.



Public Cloud:

[or, Multitenancy cloud]

A public cloud is a cloud environment (server, data center, or network) that is shared among multiple organizations. This type of cloud service is provided by a third-party vendor over the internet. This is called multitenancy, where different users (tenants) use the same physical infrastructure through virtual machines.

The advantages of Private cloud are:

- ✚ **Low Cost:** It is the most affordable cloud model. Users usually pay only for the services they use, reducing unnecessary expenses.
- ✚ **No Hardware Investment:** All infrastructure is provided and maintained by the cloud service provider.
- ✚ **No Infrastructure Management:** Users do not need to manage servers, maintenance, or IT staff, as everything is handled by the provider.

The disadvantages of Private cloud are:

- ✚ **Security and Privacy Concerns:** Since resources are shared among many users, it may be more vulnerable to security threats and data privacy risks.
- ✚ **Reliability Issues:** Public clouds may experience outages or service interruptions due to heavy user load or technical failures.
- ✚ **Limited Customization:** Users have limited control over system configuration and can only choose basic settings like OS and resource size.
- ✚ **Limited Resources:** Resources are shared among many users, so performance and scalability may be affected during high demand.

Hybrid Cloud:

The cloud computing model that combines both private and public cloud systems into 1 integrated environment is called hybrid cloud. In this model, an organization can use:

- ✚ **Private cloud** for sensitive data and critical operations
- ✚ **Public cloud** for general workloads and scalable services

The advantages of Private cloud are:

- ✚ **Cost-Effective:** Reduces overall cost by using the public cloud for most workloads and keeping only critical data in the private cloud.
- ✚ **High Security:** Sensitive data can be stored in the private cloud, while less critical data runs on the public cloud, improving data protection through separation.
- ✚ **Flexibility:** Organizations can design custom solutions based on their needs, choosing where to run each workload.

The disadvantages of Private cloud are:

- ✚ **High Complexity:** Managing and integrating multiple cloud systems (public + private + on-premises) is difficult and requires skilled management.
- ✚ **Limited Use Case:** Hybrid cloud is only useful for organizations that have both sensitive and non-sensitive workloads; otherwise, it may not be necessary or cost-effective.

Community Cloud:

A community cloud is a cloud deployment model where the infrastructure is shared by a specific group of organizations that have common goals, requirements, or concerns (such as security, policies, or regulations).

It is similar to a public cloud in terms of shared resources, but access is restricted only to selected organizations within the community.

The advantages of Private cloud are:

- ✚ **Cost Reduction:** Infrastructure costs are shared among multiple organizations, making it cheaper than a private cloud while still maintaining good performance.
- ✚ **Better Collaboration & Efficiency:** Since it is designed for a specific group or industry, it supports standardized configuration and better coordination between members.

The disadvantages of Private cloud are:

- ✚ **Shared Resources Limitations:** Resources like storage and bandwidth are shared, which may lead to performance limitations.
- ✚ **Less Popular / Limited Availability:** It is still an emerging model, so it is not widely adopted in all industries or available everywhere.

Difference Between:

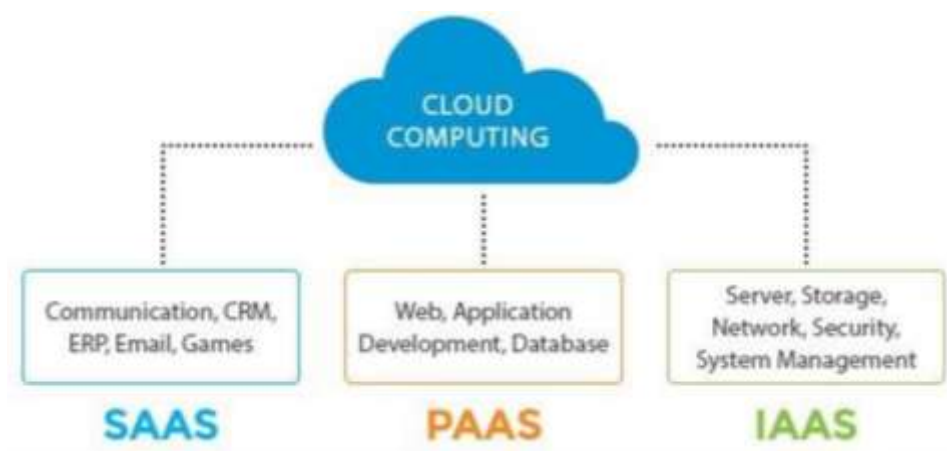
Feature	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Ease of Setup	Very easy; provider handles most setup work	Very difficult; built and managed by organization	Easy; based on shared community standards	Complex; involves integration of multiple systems
Ease of Use	Very easy to use	Complex; requires skilled in-house team	Relatively easy; members collaborate and support each other	Can be difficult depending on system integration
Data Control	Low; controlled by provider	Very high; full organizational control	Moderate to high; shared control among members	High; depends on configuration and setup
Reliability	May face outages or downtime	High reliability (if well managed)	Depends on community infrastructure	High if properly designed
Scalability	High but limited by provider plans	Very high (based on owned resources)	Moderate; depends on shared capacity	High and flexible
Security & Privacy	Low; not ideal for sensitive data	Very high; best for sensitive data	High; depends on agreed security policies	High; depends on design and integration
Flexibility	Low; fixed provider services	Very high; fully customizable	Moderate; partially standardized	Very high; highly customizable

Cost	Very low (pay-as-you-use)	Very high (infrastructure + maintenance cost)	Shared cost among members	Moderate (between public and private)
Hardware Requirement	No in-house hardware needed	Requires in-house infrastructure	Usually not required	May or may not require in-house systems

Cloud Services Models:

There are following 3 types of cloud service model:

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)



Infrastructure as a Service (IaaS):

[Hardware as a Service (HaaS)]

Infrastructure as a Service (IaaS) is a cloud service model that provides virtualized computing infrastructure over the internet. It is also known as Hardware as a Service (HaaS). In IaaS, the cloud provider manages the physical infrastructure, while users can access and use computing resources as needed.

The services provided by IaaS are:

- 🚧 **Compute services:** Provides virtual CPUs and memory for creating and running virtual machines (VMs).
- 🚧 **Storage services:** Offers online storage space for files, databases, backups, and applications.
- 🚧 **Network services:** Provides networking components such as routers, switches, and bridges for communication between VMs.
- 🚧 **Load balancing:** Distributes workloads across multiple servers to improve performance and availability.

The characteristics of IaaS are:

- 🚧 Resources are provided as online services.
- 🚧 Services are highly scalable.
- 🚧 Infrastructure is dynamic and flexible.
- 🚧 Resources can be accessed through GUI and APIs.
- 🚧 Administrative tasks are largely automated.

Examples:

Amazon Web Services (AWS), Microsoft Azure, Google Cloud, etc.

The advantages of IaaS are:

- 🚦 **Shared infrastructure:** Multiple users can share the same physical infrastructure, reducing costs.
- 🚦 **Web access to resources:** Users can access resources from anywhere through the internet.
- 🚦 **Pay-as-you-use model:** Users pay only for the resources they consume.
- 🚦 **Focus on core business:** Organizations can focus on business activities instead of managing hardware.
- 🚦 **On-demand scalability:** Resources can easily increase or decrease based on user demand.

The disadvantages of IaaS are:

- 🚦 **Security concerns:** Data and applications hosted online may face security risks.
- 🚦 **Maintenance and upgrades:** Some software upgrades and configurations may still require user management.
- 🚦 **Vendor lock-in issues:** Migrating services or virtual machines from one provider to another can be difficult.

Platform as a Service (PaaS):

Platform as a Service (PaaS) is a cloud computing model designed mainly for developers to build, test, run, and manage applications without worrying about underlying infrastructure like servers and hardware. It provides a ready-made environment where applications can be developed and deployed easily over the internet.

PaaS provides essential development components such as programming languages, application frameworks, databases, and various development tools that help in building software efficiently.

The characteristics of PaaS are:

- 🚦 Accessible to multiple users through a common development platform.
- 🚦 Easily integrates with web services and databases.
- 🚦 Built on virtualization technology, allowing resources to scale up or down as needed.
- 🚦 Supports multiple programming languages and frameworks.
- 🚦 Provides auto-scaling capability for applications based on demand.

Examples:

- 🚦 Google App Engine – PaaS platform for building and developing web applications easily.
- 🚦 AWS Elastic Beanstalk – A simple way to deploy and manage applications on AWS
- 🚦 Microsoft Azure – Widely used for enterprise level application and hosting.

The advantages of PaaS are:

- 🚦 **Simplified development:** Developers can focus on coding and innovation without managing infrastructure.
- 🚦 **Lower risk:** No need for heavy investment in hardware; only internet and a device are required.

- 🚦 **Prebuilt functionality:** Ready-made tools and services help speed up development.
- 🚦 **Strong community support:** Many platforms provide communities for learning and problem-solving.
- 🚦 **Scalability:** Applications can easily scale from small to large user bases without major changes.

The disadvantages of PaaS are:

- 🚦 **Vendor lock-in:** Applications depend on the provider's platform, making migration difficult.
- 🚦 **Data privacy concerns:** Data stored outside the organization may face security risks.
- 🚦 **Integration issues:** Combining cloud applications with local systems can be complex and may require extra effort.

Software as a Service (SaaS):

Software as a Service (SaaS) is also known as “on-demand software”. It is a cloud service model where applications are hosted and managed by a cloud provider, and users access them through the internet using a web browser without installing or maintaining the software.

In SaaS, the provider handles everything like infrastructure, updates, and maintenance, while users simply use the application.

SaaS also includes services such as business applications, document management tools, social networks, and email services.

The characteristics of SaaS are:

- 🚦 Applications are centrally managed by the provider.
- 🚦 Software is hosted on remote cloud servers.
- 🚦 Access is provided through the internet and web browsers.
- 🚦 Users do not handle updates, maintenance, or infrastructure; everything is automatic.
- 🚦 Services are usually offered on a pay-as-you-use or subscription basis.

Examples:

Gmail, YouTube, Facebook, Instagram, Zoom, etc.

The advantages of SaaS are:

- 🚦 **Easy to use and buy:** Users can start using applications quickly without setup.
- 🚦 **One-to-many model:** A single application instance is shared among many users.
- 🚦 **No hardware investment:** No need to buy or manage infrastructure.
- 🚦 **Low maintenance:** All updates and maintenance are handled by the provider.
- 🚦 **No installation required:** Access directly through browser or app.
- 🚦 **Multi-device support:** Works on laptops, phones, tablets, and more.
- 🚦 **API integration:** Easily connects with other software systems.

The disadvantages of SaaS are:

- 🚦 **Security concerns:** Data is stored in the cloud, which may raise privacy concerns for sensitive information.
- 🚦 **Latency issues:** Performance may depend on internet speed and server distance.

🌐 **Internet dependency:** Applications cannot be used without an internet connection.

🌐 **Vendor lock-in:** Switching providers can be difficult due to large data migration.

Difference Between:

Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)	Software as a Service (SaaS)
Provides virtual infrastructure like servers, storage, and networks	Provides platform + tools to develop, test, and deploy applications	Provides ready-to-use software applications
Used to build and manage virtual data centers	Used to develop and run applications	Used to directly use software over the internet
Gives access to virtual machines, storage, and networking	Gives runtime environment, frameworks, and development tools	Gives complete applications (web/software apps)
Mainly used by system/network administrators	Mainly used by developers	Mainly used by end users
User manages OS, apps, and data	User manages only applications and data	User manages nothing (only uses software)
Example: AWS EC2, Azure VMs	Example: Google App Engine, AWS Elastic Beanstalk	Example: Gmail, Zoom, Facebook

Legal Issues in Cloud computing:

Cloud computing creates several legal problems because data is stored online, shared across systems, and managed by third-party providers. The main legal issues are:

1. Data privacy and Protection:

In cloud computing, user data is stored on remote servers, so there is a risk of unauthorized access or data leaks. Cloud providers must protect this data using security measures and follow privacy laws like GDPR or local data protection rules.

2. Data Ownership:

Data ownership means who has control over the data stored in the cloud. Usually, the user owns their data, but the cloud provider stores it and manages it. Problems happen when the service agreement is not clear, because then it becomes confusing whether the provider can use the data, share it, or only store it safely for the user.

3. Jurisdiction Issues:

Cloud data can be stored in different countries. If a legal problem happens, it is difficult to know which country's law should be used. This creates confusion in legal cases.

4. Contractual Issues:

Cloud services work based on agreements (terms and conditions). Users must accept them to use the service. Problems happen if the company changes rules, deletes data, or stops service suddenly.

5. Data Security and Liability:

Data security means protecting data from hacking, loss, or misuse. In cloud computing, if data is stolen or lost, it is often unclear who is legally responsible—the user or the cloud provider. Many providers limit their responsibility in contracts, which can create problems for users.

6. Compliance Issues:

Compliance means following legal rules and regulations. Organizations using cloud services must ensure their data handling follows laws like privacy, health, or financial regulations. If the cloud provider does not meet these rules, it can lead to legal penalties or violations.

7. Intellectual Property Issues:

Intellectual property refers to ownership of digital content like documents, software, or designs. When this data is stored in the cloud, there is a risk of unauthorized copying or misuse. Legal issues occur if ownership rights are not clearly defined or protected.

Unit 2

Virtualization

Virtualization:

Virtualization is a technology that allows multiple operating systems to run on a single physical machine, where each system works independently. It hides the actual hardware and provides users with virtual resources instead. It enables one physical resource (like a server or storage) to be divided into multiple virtual resources, or multiple physical resources to be combined into one virtual system for better efficiency and management.

In virtualization, there are 2 main parts. They are:

1. **Host Machine**

The host machine is the real physical computer that provides hardware resources like CPU, memory, and storage. It runs virtualization software (hypervisor) to manage virtual machines.

2. **Guest Machine**

The guest machine is the virtual computer created inside the host machine. It runs its own operating system and applications, but it uses the resources provided by the host machine and works independently like a separate computer.

Hypervisor:

Hypervisor is a special software layer used in virtualization that sits between the hardware and Virtual machines. It manages and controls the physical resources such as CPU, memory, and storage of the host machine. It allows multiple OS to run on a single machine by creating and managing multiple virtual environments. Each works independently but are controlled by hypervisor.

Example:

Oracle VirtualVox,
Microsoft Hyper-V,
KVM (Kernal-based Virtual Machine), etc.

Types of virtualization in Cloud Computing:

1. Hardware Virtualization
2. Server Virtualization
3. Storage Virtualization
4. Data Centre Virtualization
5. OS virtualization
6. Para Virtualization

Hardware Virtualization:

Hardware virtualization is a technique in cloud computing where one physical computer is divided into multiple virtual machines. Each virtual machine works like an independent computer and can run its own operating system and applications. This makes the system more flexible and efficient. A software called a hypervisor is installed on the hardware to manage resources like CPU, memory, and storage and distribute them among virtual machines.

Hardware Virtualization is mainly of 3 types. They are:

1. **Full Virtualization:**

In full virtualization, the whole hardware is completely created in virtual form. The guest operating system runs normally without any changes, as if it is working on a real computer.

2. Emulation Virtualization:

In emulation, the virtual machine acts like real hardware. It allows different types of systems to run, even if the original hardware is different. The operating system does not need to be changed.

3. Para Virtualization:

In para-virtualization, the hardware is not fully copied. Instead, the operating system is slightly modified so it can directly communicate with the hypervisor. This makes the system faster and more efficient.

The advantages of hardware virtualization are:

- ✚ **Low cost:** One physical machine can run many virtual machines, so less hardware is needed.
- ✚ **Better resource use:** CPU, memory, and storage are shared efficiently among virtual machines.
- ✚ **Easy management:** Virtual machines can be created, deleted, or changed easily.
- ✚ **Flexibility:** Different operating systems and applications can run on the same hardware.
- ✚ **High efficiency:** Resources are used properly without wasting hardware power.
- ✚ **Live migration support:** A running virtual machine can be moved to another server without stopping it.
- ✚ **Improved scalability:** More virtual machines can be added easily when needed.

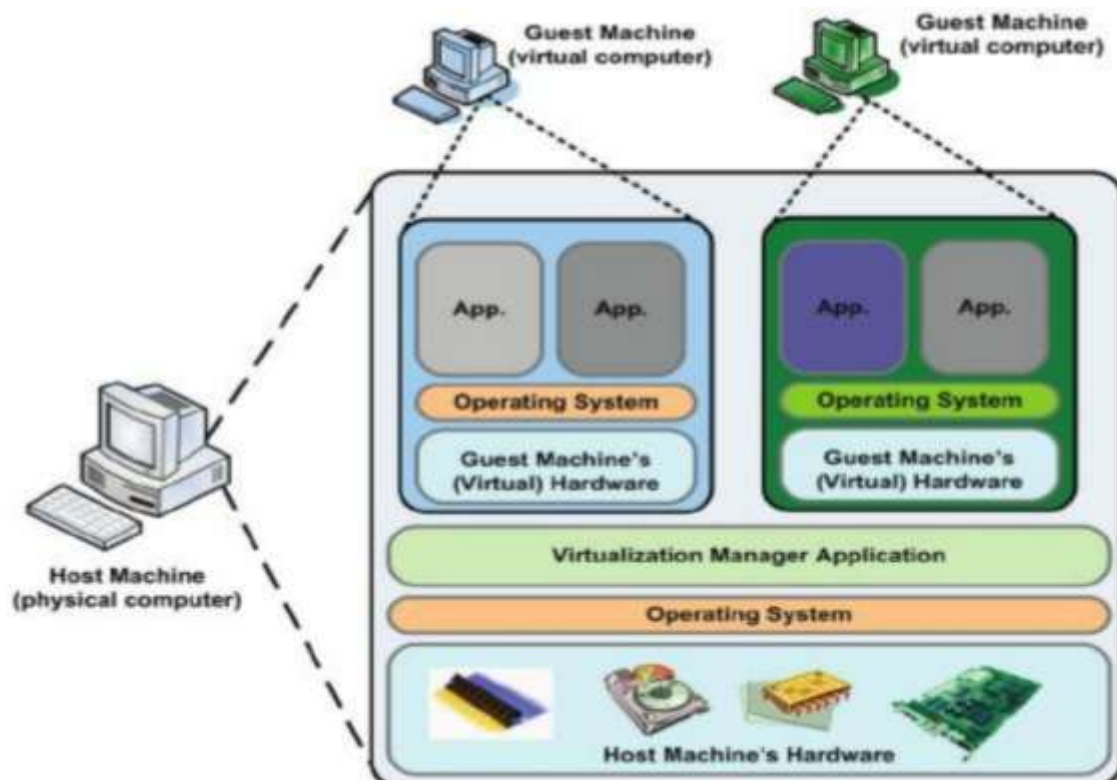


Fig: Hardware virtualization

Server Virtualization:

Server virtualization is a process in which a single physical server is divided into multiple virtual servers using virtualization software. This is done using a hypervisor software. Each virtual server works independently and can run its own operating system and applications. This improves resource utilization and reduces the need for multiple physical machines.

The advantages of Server Virtualization:

- ✚ **Cost saving:** Reduces the need for multiple physical servers, lowering hardware and maintenance costs.
- ✚ **Better resource utilization:** CPU, memory, and storage are used efficiently by sharing them among virtual servers.
- ✚ **Easy scalability:** New virtual servers can be created quickly when demand increases.
- ✚ **Improved flexibility:** Different operating systems and applications can run on the same physical server.
- ✚ **Energy saving:** Fewer physical servers means less electricity and cooling cost.
- ✚ **Easy backup and recovery:** Virtual servers can be quickly backed up and restored.
- ✚ **High availability:** If one virtual server fails, others continue working without interruption.
- ✚ **Simplified management:** Servers can be monitored and controlled more easily from one place.

The disadvantages of Server Virtualization:

- ✚ **Performance overhead:** Virtual machines may run slower because resources are shared through a hypervisor.
- ✚ **Single point of failure:** If the main physical server fails, all virtual machines on it can be affected.
- ✚ **High initial setup cost:** Powerful hardware and virtualization software are needed at the start.
- ✚ **Complex management:** Managing many virtual machines can become difficult and require skilled administrators.
- ✚ **Security risks:** If one virtual machine is attacked, there is a risk it may affect other systems on the same server.
- ✚ **Resource contention:** Multiple virtual machines may compete for CPU, memory, and storage, reducing performance.

Data Centre Virtualization:

Data Centre Virtualization is the process of converting the physical resources of a data center (such as servers, storage, and networks) into virtual resources that can be managed and used more efficiently. Instead of depending on separate physical machines for each task, virtualization creates a pool of resources and divides them into multiple virtual machines and virtual systems as needed. This allows better utilization of hardware, reduces cost, improves flexibility, and makes it easy to scale and manage the entire data centre from a single system.

The advantages of Data Centre virtualization:

- ✚ **Better resource use:** Hardware is shared efficiently, so nothing stays idle.
- ✚ **Cost saving:** Fewer physical servers are needed, reducing hardware and electricity costs.
- ✚ **Easy management:** Systems can be controlled and monitored from one place.
- ✚ **Scalability:** New virtual machines can be created quickly when needed.
- ✚ **Flexibility:** Different operating systems and applications can run on the same hardware.
- ✚ **Backup & recovery:** Data and systems can be restored easily using snapshots.

The disadvantages of Data Centre virtualization:

- ✚ **High initial setup cost:** Requires powerful servers and software.
- ✚ **Performance overhead:** Virtual machines may run slightly slower than physical machines.
- ✚ **Single point of failure:** If the main physical host fails, multiple VMs are affected.
- ✚ **Security risks:** If not managed properly, virtual environments can be vulnerable.
- ✚ **Complex management:** Needs skilled administrators to handle virtualization tools.

OS Virtualization:

[or, System level virtualization]

Operating System Virtualization is a type of virtualization where multiple isolated environments (called containers) run on a single operating system kernel. Instead of creating full virtual machines, it shares the same OS but separates applications so they behave like independent systems.

Example: Docker

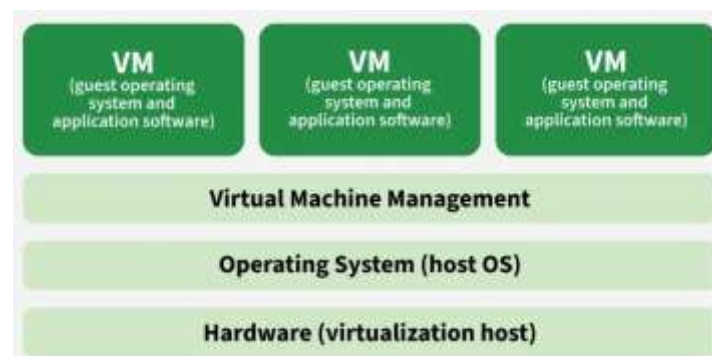


Fig: OS virtualization

The uses of OS Virtualization:

- ✚ It is used to run many applications separately on one computer without conflict
- ✚ It helps developers test software safely without affecting the main system
- ✚ It is used in web hosting to run multiple websites on a single server
- ✚ It supports cloud systems by quickly creating application environments
- ✚ It is used in microservices to run different parts of an application independently
- ✚ It improves system usage by sharing CPU, memory, and storage efficiently

The advantages of OS Virtualization:

- ✚ **Lightweight:** Uses less memory and CPU than full virtual machines
- ✚ **Fast performance:** No need to run separate operating systems
- ✚ **Efficient resource use:** Many apps can run on one OS
- ✚ **Quick setup:** Containers can be created and started very quickly
- ✚ **Scalable:** Easy to run many instances of an application

The disadvantages of OS Virtualization:

- ✚ **Same OS limitation:** All containers must use the same operating system kernel
- ✚ **Weaker isolation:** Less secure compared to full virtual machines
- ✚ **Kernel dependency:** If OS kernel fails, all containers are affected
- ✚ **Not suitable for all workloads:** Some applications need full OS control

Note: Hypervisor is not used for OS virtualization.

Para Virtualization:

Paravirtualization is a type of virtualization where the guest operating system is aware that it is running inside a virtual machine instead of on real hardware. In this method, the hypervisor and the guest operating system work together. The guest OS does not try to act like it is using real physical hardware; instead, it directly communicates with the hypervisor using special requests called hypercalls.

This cooperation reduces unnecessary processing and makes the system faster and more efficient compared to full virtualization. In simple terms, paravirtualization means the operating system knows it is virtual and interacts directly with the virtual system to improve performance.

The advantages of OS Virtualization:

- ✚ Faster performance (less overhead)
- ✚ Efficient use of system resources
- ✚ Direct communication with hypervisor
- ✚ Better scalability in virtual environments

The disadvantages of OS Virtualization:

- ✚ Not all operating systems are supported
- ✚ Guest operating system must be modified to run
- ✚ Less flexible than full virtualization
- ✚ Difficult to implement compared to other virtualization types

Role of Virtualization in enabling Cloud Services:

Virtualization plays a very important role in enabling cloud services. Most benefits of cloud computing like easy scaling, flexible resources, and better security are possible because of virtualization. It allows cloud providers to quickly create new virtual servers whenever needed, instead of setting up physical machines. If more memory or resources are required, they can be easily added to an existing virtual server without changing hardware. In this way, virtualization makes it possible to use computing resources on demand, and it becomes the main technology behind cloud computing.

Difference Between:

Cloud Computing	Virtualization
It is a service that provides computing resources (like servers, storage, software) over the internet.	It is a technology that creates virtual versions of physical hardware.
Users access resources online from cloud providers.	It is used inside systems to divide physical resources into virtual machines.
Works on a large scale (global access).	Works on a single machine or data center level.
Focus is on delivering services to users.	Focus is on creating virtual resources from hardware.
Depends on virtualization.	It is the base technology used to build cloud computing.
Example: Google Drive, AWS, Azure.	Example: Virtual machines using KVM or Hyper-V.

Unit 3

Cloud Migration

Cloud Migration:

Cloud migration is the process of moving digital business operations into the cloud. It involves transferring data, applications, and IT processes from on-premises or legacy data centres to cloud-based data centres. This process is similar to moving to a new office, but instead of physical items, digital resources are shifted.

Cloud migration requires proper planning and preparation, but it is usually worth the effort as it provides benefits like cost savings and greater flexibility. Although it mostly refers to moving from on-premises systems to the cloud, it can also mean migrating from one cloud service to another.

Types of Cloud Migration Strategies:

The main types of cloud migration strategies are:



Rehosting:

This means moving applications from on-premises to the cloud without making major changes. It is the fastest and simplest method.



Replatforming:

In this strategy, small optimizations are made before moving to the cloud, but the core system remains the same.



Refactoring / Re-architecting

This involves redesigning the application to fully use cloud features. It gives the best performance but takes more time and effort.



Repurchasing

Here, the existing application is replaced with a cloud-based software (SaaS), such as switching to a cloud CRM system.



Retiring

This involves identifying and removing applications that are no longer useful or needed, helping to reduce cost and complexity.



Retaining

This means keeping certain applications in the existing on-premises system because they are not suitable or ready for cloud migration.

Types of Cloud Migration:

The types of Cloud Migration are:

1. Shallow Cloud Migration
2. Deep Cloud Migration

Shallow Cloud Migration

Shallow cloud migration, also known as rehosting, is the process of moving applications and workloads from one environment (usually on-premises) to a cloud environment without changing the application architecture or code. It simply transfers systems from one location to another, making it a fast and simple migration method.

Because there are little or no changes to the application, this approach is less time-consuming and less complex compared to other migration strategies. It also allows organizations to quickly move from capital expense (CapEx) to operational expense (OpEx), helping to start a hybrid cloud setup.

The benefits of shallow Cloud Migration are:

- ✚ Cost-effective and quick to implement with no major application changes
- ✚ Security model remains unchanged with no major new risks introduced
- ✚ Removes hardware limitations by enabling easy scaling in the cloud
- ✚ Provides higher reliability and better system availability
- ✚ Serves as a good starting point for future deeper cloud migration

Deep Cloud Migration:

Deep cloud migration is the process of moving applications to the cloud by modifying and redesigning them to fully use cloud capabilities. Unlike simple migration, the application is improved during the migration process to take advantage of features like auto-scaling, dynamic load balancing, and even serverless computing.

This approach is more complex and time-consuming because it requires significant changes to the system. However, it provides better performance, efficiency, and long-term benefits. Some organizations also implement deep cloud integration in parts of their system to make the process easier, although this can still take more time and resources.

The benefits of Deep Cloud Migration are:

- ✚ Faster deployment of applications and services
- ✚ Improved employee collaboration through better access to data and applications
- ✚ Strong disaster recovery with secure data backup and protection from failures or disasters
- ✚ Enables edge computing for low-latency and high-performance applications

Need for cloud Migration:

Cloud migration is needed because of following reasons:

- ✚ Reduce IT infrastructure and maintenance cost
- ✚ Easily scale up/down resources based on demand
- ✚ Access data and applications from anywhere using the internet
- ✚ Improve performance and efficiency
- ✚ Enhance data security, backup, and disaster recovery
- ✚ Increase system reliability and reduce downtime
- ✚ Enable Faster deployment and updates of applications
- ✚ Replace outdated legacy systems with modern cloud solutions

Model of Migration into a Cloud / Process of Migrating Cloud:

Cloud migration is a complex process, so it follows a step-by-step model to make it smooth, organized, and error-free. The main steps are:

1. Assess

In this step, the current IT environment is analyzed. Organizations study existing applications, data, workloads, and infrastructure to understand what can be moved to the cloud. They also identify dependencies, risks, and business requirements.

2. Plan

After assessment, a detailed migration plan is created. This includes choosing migration strategy, estimating cost, setting timelines, identifying required resources, and defining possible risks and mitigation methods.

3. Design

In this stage, the target cloud environment is designed. Decisions are made about cloud deployment model (public, private, hybrid), service model (IaaS, PaaS, SaaS), security setup, network architecture, and data storage structure.

3. Migrate

This is the actual process of moving applications, data, and workloads from the existing system to the cloud. Migration is done carefully to avoid downtime and data loss.

4. Validate

After migration, the system is tested to ensure everything is working properly. Performance, security, data accuracy, and application functionality are checked against requirements.

5. Optimize

Once the system is running in the cloud, it is improved for better performance and cost efficiency. This may include scaling resources, improving speed, and reducing unnecessary usage.

6. Govern

In the final stage, continuous monitoring and management are done. Security policies, compliance rules, cost control, and system performance are regularly checked to ensure smooth cloud operations.

Various Risks in Cloud Migrating:

The various risks involved in cloud migrating are:

-  **Far less user control:** Users do not have full control over cloud infrastructure, and failures on provider-side systems can affect services and data.
-  **Unauthorized instance spawning:** If login credentials are stolen, attackers can create new cloud instances, increasing cost and potentially accessing sensitive data.
-  **API vulnerabilities:** Cloud services depend on APIs, and weak security in APIs can allow attackers to access or manipulate data.
-  **Shared cloud environment risks:** Since multiple users share cloud infrastructure, poor isolation may lead to data exposure between tenants.
-  **Secure deletion issues:** Completely deleting data is difficult in cloud systems, which may leave residual data accessible.
-  **Improper user privileges:** Weak access control or shared accounts can allow unauthorized users to misuse resources.
-  **Vendor lock-in:** Switching from one cloud provider to another can be difficult, expensive, and time-consuming.

- ✚ **Overworked IT staff:** Cloud migration increases workload, which may lead to mistakes and system mismanagement.
- ✚ **Insider threat:** Authorized users with malicious intent can damage systems or leak sensitive data.
- ✚ **Data loss:** Hardware failures, attacks, or disasters may lead to unexpected loss of important data.
- ✚ **Multiple suppliers issue:** Using many cloud providers increases complexity and reduces control over data flow.
- ✚ **Lack of proper research:** Poor planning and insufficient understanding before migration can lead to serious errors and system failure.

Unit 4

Software Quality

Communication as a Service (CaaS):

Communication as a Service (CaaS) is a cloud service model that provides communication tools and services over the internet. Instead of installing and managing communication systems in an organization, users can access them through the cloud. CaaS includes services like voice calls, video conferencing, messaging, email, and collaboration tools. These services are provided by a third-party provider and can be used from anywhere using an internet connection.

It helps organizations reduce the cost of setting up communication infrastructure and makes communication easier, faster, and more flexible. Users can easily scale communication services based on their needs without maintaining hardware or software.

Example: (Tools)

Zoom, Microsoft Teams, Google Meet, etc.

The benefits of CaaS are:

-  **Low cost:** Reduces the need for expensive communication hardware and setup
-  **Easy communication:** Enables voice, video, and messaging through a single platform
-  **Remote access:** Users can communicate from anywhere using the internet
-  **Scalability:** Easily add or remove users based on business needs
-  **Better collaboration:** Improves teamwork through real-time communication tools
-  **No maintenance:** The service provider handles updates, security, and system management

Monitoring as a Service (MaaS):

Monitoring as a Service (MaaS) is a cloud service that helps organizations monitor their IT systems, applications, and networks using the cloud. Instead of installing and managing monitoring tools on their own computers or servers (on-premises), companies use a cloud provider's monitoring system. This makes the process easier, faster, and cheaper.

MaaS is useful because it reduces the need for physical infrastructure and IT staff. It also allows real-time monitoring of performance, security, and system health from anywhere. In traditional on-premises monitoring, companies must buy software, set up hardware, and hire skilled staff to manage everything. This is costly and difficult to maintain. MaaS removes most of this burden by providing monitoring as a ready-made cloud service.

Example: (Tools)

Amazon CloudWatch, Microsoft Azure Monitor, Google Cloud Monitoring, etc.

The benefits of MaaS are:

-  **Ready-to-use monitoring system:** The service provider sets up the hardware, monitoring tools, configurations, and alerts, so users can start using it easily without setup effort.
-  **24x7 availability:** MaaS is cloud-based, so the monitoring dashboard is always available and can be accessed anytime from anywhere.
-  **Easy integration with business processes:** MaaS can be easily integrated with business systems. It sends alert messages based on important business events, helping companies respond quickly to issues.

- 🌐 **Cloud compatible:** MaaS works smoothly with other cloud services like SaaS and PaaS, making integration easier in cloud environments.
- 🌐 **No maintenance required:** Users do not need to manage infrastructure or monitoring systems, as everything is handled by the service provider.

Jericho Cloud Cube Model:

The Jericho Cloud Cube Model is developed by the Jericho Forum to help organizations understand and evaluate cloud security and cloud environments. It is used to decide how safe and suitable a cloud setup is for an organization.

This model explains cloud computing using four key dimensions, which help classify different types of cloud systems and their security level. The four dimensions are:

1. Physical Location of Data

This shows where the data is stored – either inside the organization (internal/on-premises) or outside the organization (external/cloud provider). It helps define the security boundary of the organization.

2. Ownership

This refers to who owns and controls the cloud system. It can be proprietary (owned by a company) or open (shared or community-based systems). It also considers how easily data can move between systems and the level of vendor lock-in.

3. Security Boundary

This describes whether operations happen inside a secure boundary (protected environment like firewall) or outside it (less controlled environment). It helps measure security exposure.

4. Sourcing

This defines who provides the service – either in-sourcing (managed internally by the organization) or out-sourcing (managed by a cloud service provider).

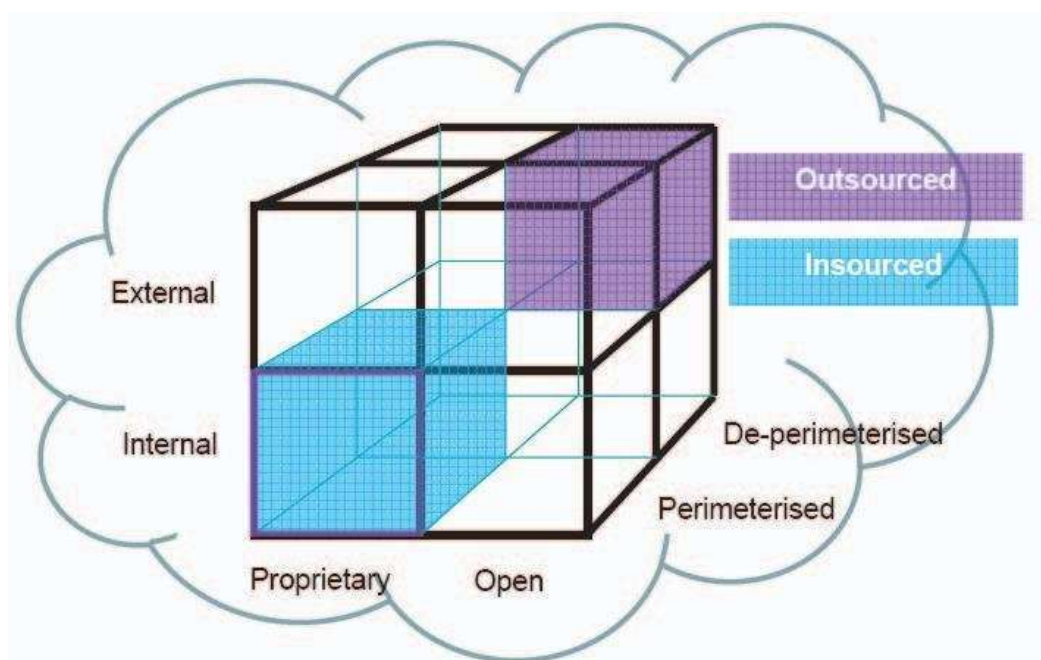


Fig: Cloud Cube Model

Amazon's Web Services (AWS):

- ✚ Amazon Web Services (AWS) was launched in 2006 by Amazon to provide IT services over the internet in the form of web services, which is now known as cloud computing.
- ✚ With AWS, organizations do not need to plan or maintain physical servers and other IT infrastructure in advance. Instead, they can quickly launch hundreds or even thousands of servers within minutes, making systems faster and more scalable.
- ✚ AWS follows a **pay-as-you-go model**, meaning users only pay for the services they use. There are no upfront costs or long-term commitments, which makes it cost-effective and flexible.
- ✚ AWS provides a wide range of cloud services such as computing power, storage, databases, networking, and many other tools needed to build and run applications.

The web services provided by AWS are as follows:

1. Amazon Elastic Cloud Compute (EC2)
2. Amazon S3 (Simple Storage Service)
3. Amazon Virtual Private Cloud (VPC)
4. Amazon CloudFront
5. Amazon Relational Database Service (RDS)

Amazon Elastic Cloud Compute (EC2):

- ✚ Amazon Web Services EC2 is a cloud computing service provided by AWS that allows users to create and use virtual servers in the cloud.
- ✚ It helps organizations reduce the need for expensive physical servers by providing virtual machines that can be created and managed through a web interface. Users can quickly launch servers with their preferred operating system in just a few minutes.
- ✚ Amazon EC2 provides resizable compute capacity, meaning users can easily increase or decrease server resources based on their needs. It also offers features for managing security, storage, and network settings of virtual servers.
- ✚ EC2 makes server management easier, allowing users to focus more on application development and projects instead of maintaining physical infrastructure.

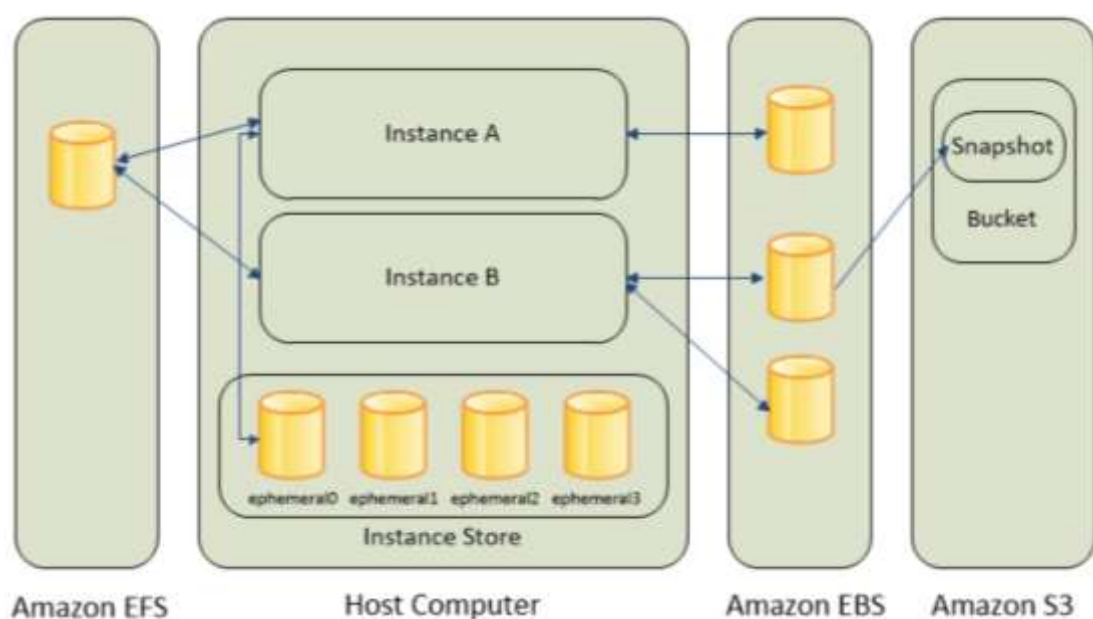


Fig: Amazon Elastic Cloud Compute (EC2)

The characteristics of Amazon EC2 (Elastic Cloud Compute) are:

- ✚ **Elastic (Scalable):** Resources can be increased or decreased easily based on demand.
- ✚ **On-demand service:** Users can launch and use virtual servers anytime without prior setup.
- ✚ **Pay-as-you-go:** You only pay for the compute resources you use.
- ✚ **Flexible configuration:** Users can choose operating system, storage, memory, and CPU as needed.
- ✚ **Quick deployment:** Virtual machines (instances) can be created and started within minutes.
- ✚ **Secure:** Provides security features like firewall (security groups) and network control.
- ✚ **Reliable:** Offers high availability and runs on Amazon's global infrastructure.
- ✚ **Managed access:** Users can control instances remotely and manage them easily.

The services of Amazon EC2 are:

- ✚ **EC2 Instances:** Virtual servers used to run applications and workloads in the cloud.
- ✚ **Amazon Machine Images (AMIs):** Pre-built templates used to quickly launch EC2 instances.
- ✚ **Elastic Block Store (EBS):** Provides storage for EC2 instances.
- ✚ **Elastic Load Balancing (ELB):** Distributes traffic across multiple instances to improve performance.
- ✚ **Security Groups:** Acts as a virtual firewall to control access to instances.
- ✚ **Key Pairs:** Used for secure login to EC2 instances.
- ✚ **Elastic IP Address:** A fixed public IP address that can be assigned to an instance.

Amazon S3 (Simple Storage Service):

Amazon Web Services S3 is a cloud storage service provided by AWS that allows users to store and retrieve data over the internet. Amazon S3 stores data in the form of objects inside storage buckets. It is mainly used for storing files, backups, images, videos, documents, and application data securely in the cloud.

S3 provides high security, durability, and availability by storing data across multiple physical locations and data centres. This helps protect data from loss and ensures reliable access at anytime from anywhere through the internet. It also allows users to quickly upload, manage, and retrieve data with very low access time, making it suitable for both personal and business use.

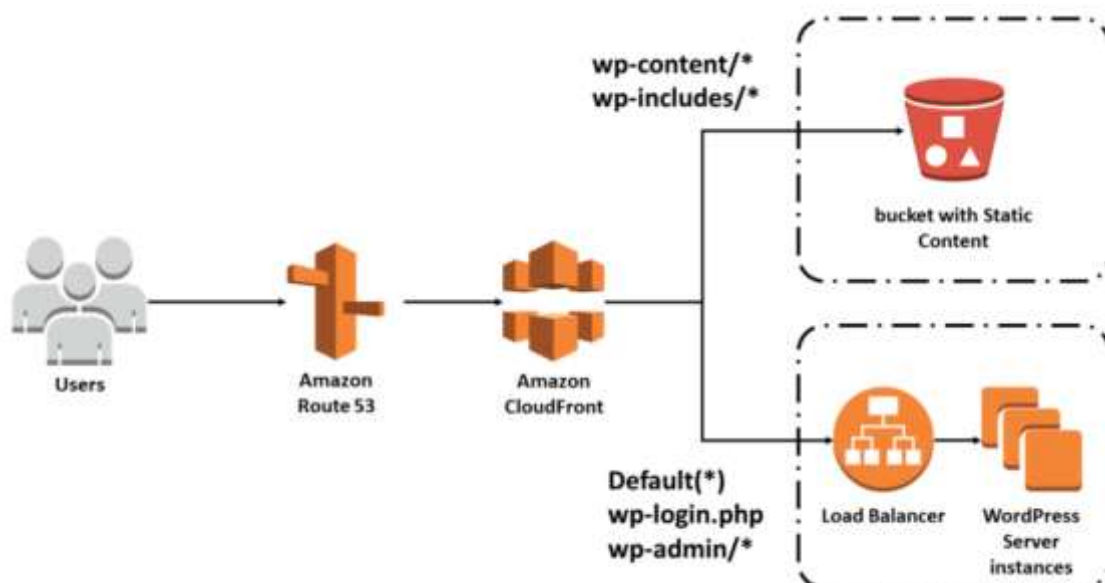
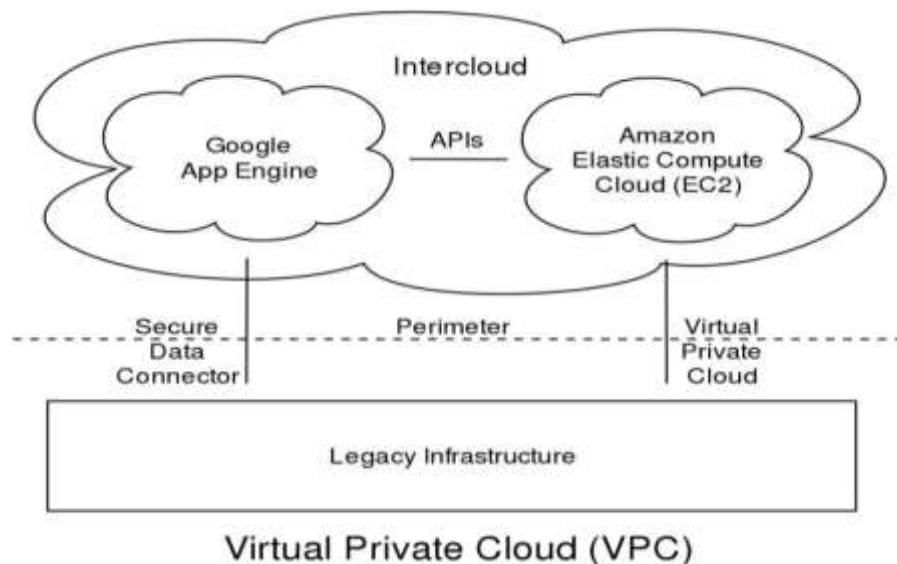


Fig: Representation of Amazon S3 and Amazon CloudFront

Amazon Virtual Private Cloud (VPC):

Amazon Web Services Amazon VPC is a networking service in AWS that allows users to create a private virtual network within the AWS cloud. It helps organizations securely run their cloud resources in an isolated network environment. Each AWS account can create its own virtual network, which is logically separated from other networks in the cloud. Amazon VPC provides control over network settings such as IP addresses, subnets, routing, and security rules. This isolation and security help protect user data and resources from unauthorized access, making cloud infrastructure safer and more reliable.



Amazon CloudFront:

Amazon Web Services Amazon CloudFront is a content delivery service in AWS that helps deliver web content to users with high speed and low latency. It distributes content such as websites, videos, images, and applications through a global network of servers located in different regions.

CloudFront works smoothly with other AWS services and helps developers deliver content to end users quickly and efficiently. By storing cached copies of content closer to users, it reduces loading time and improves performance, making content delivery faster and more reliable.

Amazon Relational Database Service (RDS):

Amazon Web Services Amazon RDS is a database service in AWS that helps users create, manage, and operate relational databases in the cloud. It is designed to handle database-related workloads and store structured data efficiently.

RDS supports several popular database systems such as MySQL, Oracle, Microsoft SQL Server, and MariaDB. It simplifies database management by automating tasks like backups, updates, maintenance, and scaling.

By reducing the need for manual database administration, Amazon RDS lowers operational costs and allows users to focus more on applications instead of database maintenance.

Cloud Computing from Google Perspective:

Google Cloud Platform (GCP) is a public cloud service that provides computing resources over the internet on a pay-as-you-go basis. It helps users build, deploy, and manage applications without needing physical infrastructure.

GCP uses Google's global infrastructure to run applications and deliver services worldwide. It supports both virtual machines and modern cloud-native, container-based applications, making development faster and more efficient.

It also offers flexible pricing with discounts for long-term and continuous usage, helping businesses reduce costs.

The benefits of GCP are:

-  Cost-effective with flexible pricing models
-  Easy and fast application deployment
-  Supports modern cloud-native applications
-  High scalability using Google's global infrastructure
-  Reliable and secure cloud environment

Windows Azure and Online services:

Microsoft Azure is a public cloud computing platform developed by Microsoft. It provides a wide range of cloud services including Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS), allowing users to build, deploy, and manage applications in the cloud.

Azure offers more than 600 cloud services and supports various operating systems, databases, programming tools, and frameworks. This makes it a flexible and powerful platform for developing and running applications across different environments.

The Microsoft Azure Services are:

1. Azure Application Services
2. Azure Data Services
3. Azure Development Services
4. Azure Compute Services
5. Azure Network Services

Azure Application Services:

Azure Application Services provide tools to help develop, deploy, and manage applications on the Microsoft Azure platform. They make it easier for developers to build and run applications without managing complex infrastructure.

It includes tools like Azure AI, Azure Analytics, Azure Internet of Things (IoT), Azure Active Directory (AD), Azure Media Services, Azure Scheduler.

Azure Data Services

Azure Data Services provide tools to store, manage, and process data in the Microsoft Azure cloud. They help organizations handle large amounts of data securely and efficiently for applications and business needs.

It includes tools like Azure Storage, Azure SQL Database, Azure Cosmos DB, Azure Data Lake, Azure Synapse Analytics.

Azure Development Services

Azure Development Services provide tools that help developers build, test, and deploy applications in the cloud. They make software development faster and easier by supporting coding, teamwork, and automation.

It includes tools like Azure DevOps, Azure Repos, Azure Pipelines, Azure Boards, Azure Test Plans.

Azure Compute Services:

Azure Compute Services provide the computing power needed to run applications and workloads in the Microsoft Azure cloud. They allow users to run programs, host applications, and process data without needing physical servers.

It includes tools like Azure Virtual Machines (VMs), Azure Container Instances, Azure Kubernetes Service (AKS), Azure App Services.

Azure Network Services

Azure Network Services provide tools to connect, secure, and manage communication between cloud resources and users over the internet. They help ensure fast, reliable, and secure data transfer in the Azure cloud.

It includes tools like Azure Content Delivery Network (CDN), Azure Load Balancer, Azure VPN Gateway.

Unit 5

SOA and Cloud

Service Oriented Architecture (SOA):

Service-Oriented Architecture (SOA) is a design approach where applications are built using independent services available over a network. These services work together to form complete applications. In SOA, different services communicate with each other over the internet or network to perform specific tasks and deliver results.

There are 2 main roles in Service Oriented Architecture. They are:

1. Service Provider:

The service provider is the organization that creates and maintains services. It makes one or more services available for others to use. The provider can publish these services in a registry with a service contract. This contract explains what the service does, how to use it, any requirements, and possible costs or fees.

2. Service Consumer:

The service consumer is the user or application that uses available services. It finds service information from the registry and connects to the required service. The consumer may also combine multiple services or use workflows to complete a task or meet a specific need.

Consumers Layer

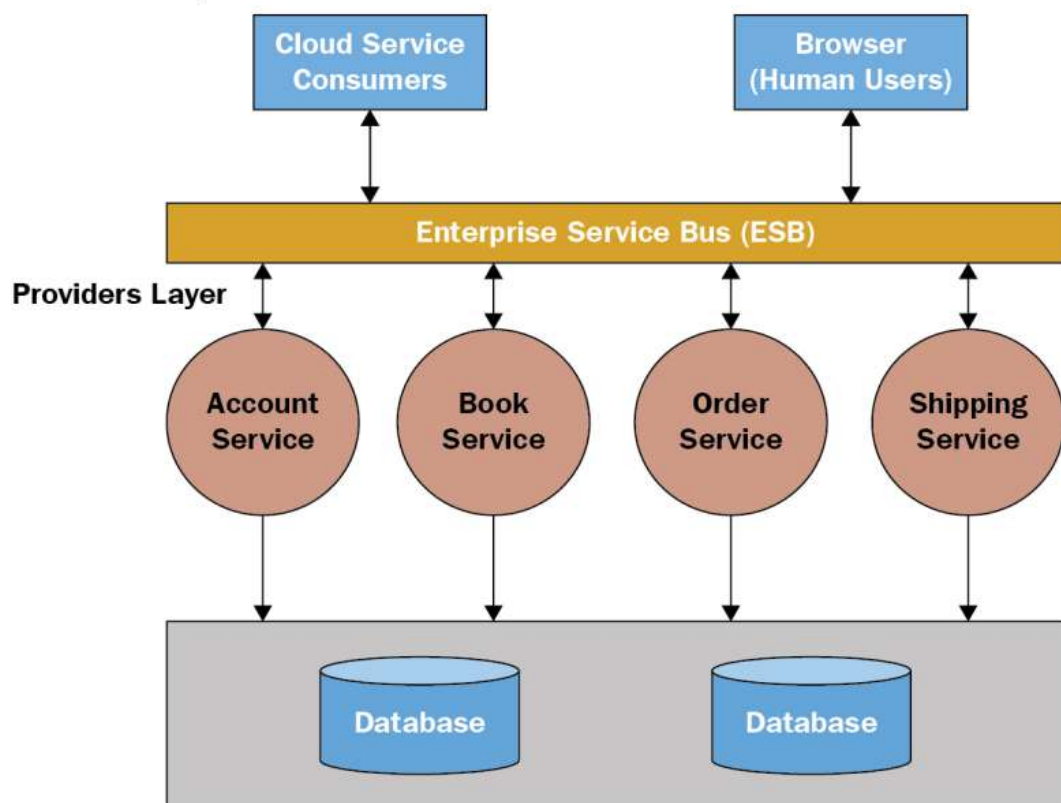


Fig: Service Oriented Architecture

Guiding Principles of SOA:

[also, Principle Characteristics of SOA]

The guiding principles of SOA (Service-Oriented Architecture) are the basic rules or ideas that explain how SOA should be designed and used. The principles of SOA are given as:

-  **Standardized Service Contract:**
Each service is defined using a standard contract or service description document that explains how it works and how it can be used.
-  **Loose Coupling:**
Services are designed to be independent, with minimal dependency on other services, so changes in one service do not affect others.
-  **Abstraction:**
Internal working of services is hidden; only service interface is visible to users.
-  **Reusability:**
Services are designed in a way that they can be reused in different applications, which reduces development time and cost.
-  **Autonomy:**
Each service controls its own logic and runs independently without needing to expose its implementation.
-  **Discoverability:**
Services are described with metadata so they can be easily found and used when needed, including third-party services.
-  **Composability:**
Multiple services can be combined together to build complex applications.

Advantages of SOA:

-  **Service reusability:** Existing services can be reused to build multiple applications, saving time and effort.
-  **Easy maintenance:** Each service works independently, so it can be updated or changed without affecting others.
-  **Platform independent:** Services can work together even if they are built on different platforms or technologies.
-  **Availability:** Services are easily accessible and can be used whenever they are needed.
-  **Reliability:** Smaller independent services are easier to test and debug, making the system more reliable.
-  **Scalability:** Services can run on multiple servers, allowing the system to handle increased load easily.

Disadvantages of SOA:

-  **High overhead:** Each service interaction requires validation of inputs, which increases processing time, load, and reduces performance.
-  **High investment:** SOA requires significant initial cost for setup, tools, and infrastructure.
-  **Complex service management:** Services communicate through many messages, which can become very large in number and difficult to manage and monitor.

Q. SOA is a black-box component architecture. What do you mean by it?

It means that the internal working (implementation details) of each service is hidden from the user. Users can only access and use the service through its interface or service contract without knowing how it works inside. It is also called abstraction.

Combining SOA with Cloud Computing:

Combining SOA (Service-Oriented Architecture) with cloud computing means using SOA principles to design cloud-based services in a more structured and efficient way. SOA breaks applications into independent and reusable services, while cloud computing provides the infrastructure to run these services with scalability, flexibility, and high availability. When both are combined, businesses can easily build, deploy, and manage applications in the cloud.

SOA helps organize applications into reusable services, making it easier to move them to cloud environments. Cloud platforms then provide the resources needed to run these services efficiently. This combination improves system flexibility, reduces cost, increases reuse of services, and makes it easier to update and manage applications.

Importance of SOA in cloud Computing:

The importance of SOA in Cloud Computing is given as:

- ✚ SOA hides internal implementation details, so users can use services without knowing the programming language behind them.
- ✚ SOA and cloud computing work together and support each other well.
- ✚ SOA makes it easier for businesses to move and integrate applications into the cloud.
- ✚ It helps in easy deployment of SaaS applications using cloud resources like storage and processing power.
- ✚ It improves communication and collaboration between different systems and services.
- ✚ Cloud provides infrastructure, while SOA provides the structure to organize and connect services.
- ✚ SOA allows reuse of existing services, which reduces cost and saves development time.
- ✚ Reusable services reduce extra effort in testing and maintenance.
- ✚ SOA components are independent, so they can easily work with cloud systems and scale efficiently.

Unit 6

Cloud Security

Cloud Security:

Cloud security means protecting cloud-based systems, data, applications, and services from unauthorized access, attacks, loss, or damage. For example, when we store files in Google Drive, the cloud provider uses security method to protect our information,

The main goal of Cloud security:

-  **Confidentiality:** Keeping data private so that only authorized users can access it.
-  **Integrity:** Ensures data is consistent and unchanged once stored.
-  **Availability:** Data and cloud services should be accessible whenever needed.

Cloud Security Challenges:

Cloud security challenges are the problems and risks faced while protecting data, applications, and services in cloud computing.

The major cloud security challenges are explained below:

-  **Data Breach**
A data breach occurs when unauthorized users gain access to sensitive cloud data. This can lead to financial loss, legal issues, and damage to trust and reputation.
-  **Misconfiguration (Wrong Settings)**
Misconfiguration happens when cloud services are set up incorrectly, making private data or resources publicly accessible. It is one of the most common causes of cloud security issues.
-  **Weak Security Planning**
This occurs when organizations move to the cloud without a proper security strategy. Lack of planning leads to poor protection and increases the risk of cyber-attacks.
-  **Weak Access Control (IAM Issues)**
If identity and access management is not properly handled, unauthorized users may gain access to cloud resources. Weak passwords and incorrect permissions are major risks.
-  **Account Hacking (Hijacking)**
Account hijacking happens when attackers steal login credentials and take control of cloud accounts. This can result in data theft, misuse of services, and system damage.
-  **Insider Threat**
Insider threats come from employees or trusted users who intentionally or accidentally expose or misuse sensitive data. These threats are often hard to detect.
-  **Unsafe APIs**
Cloud services use APIs to communicate and manage resources. If APIs are poorly designed or unsecured, attackers can exploit them to access or manipulate data.
-  **Weak Cloud Control System**
A weak control system means cloud resources are not properly monitored or managed. This can lead to security gaps, data leakage, or system failure.

System or Infrastructure Failures

Failures in the cloud provider's internal systems or infrastructure can affect security and service availability, sometimes leading to data loss or downtime.

Poor Monitoring of Cloud Use

When organizations cannot properly track cloud activity, unauthorized or risky usage may go unnoticed. This increases the chance of security incidents.

Misuse of Cloud Services

Attackers may misuse cloud platforms to host malware, launch attacks, or perform illegal activities, taking advantage of trusted cloud infrastructure.

Cloud Security Dimensions:

Cloud security has different important “dimensions” (methods or areas) used to protect cloud systems, data, and applications. The main dimensions are explained below in a simple and clear way: The dimensions are:

1. Cloud Firewalls
2. Penetration Testing
3. Data Obfuscation

Cloud Firewalls:

Cloud firewalls are software-based security systems used in the cloud to control network traffic and block unauthorized access to private networks. They monitor incoming and outgoing data and apply security rules to allow or block traffic based on safety conditions.

The benefits of cloud firewalls are:

-  **Scalability:** Can easily increase or decrease protection as network size grows.
-  **High availability:** Built with backup systems to ensure continuous protection even if one system fails.
-  **Remote access:** Can be installed and managed from anywhere through the internet.
-  **Traffic filtering:** Can control traffic between internet, virtual networks, and cloud environments.
-  **Strong security:** Provides protection similar to traditional (on-premise) firewalls.
-  **Access control integration:** Works with user access systems for better security control.
-  **Monitoring & logging:** Helps track network activity and detect threats easily.

Penetration Testing:

Penetration testing is the process of testing a cloud system, network, or application to find security weaknesses before hackers can exploit them. It is done by ethical hackers who try to break into the system safely.

The benefits of Penetration testing are:

-  Identifies security vulnerabilities
-  Improves system protection
-  Helps prevent real cyber attacks
-  Keeps security systems updated and strong

Data Obfuscation:

Data obfuscation refers to the techniques used to hide or protect sensitive data so that even if it is stolen, it cannot be easily understood. It makes data useless to attackers while keeping it safe for organizations.

The common methods / techniques for Data Obfuscation are:



Encryption

Encryption is a method of converting normal readable data into an unreadable form using a secret key. Only authorized users with the correct key can convert it back to readable form. It is mainly used to protect data during storage and transmission.



Tokenization

Tokenization is the process of replacing sensitive data with a random value called a token. The token has no meaningful value outside the system. The original data is stored securely and can only be accessed by authorized users when needed.



Data Masking

Data masking is a technique where real data is replaced with fake but realistic-looking data. It is used so that users can work with data (like testing or training) without exposing the actual sensitive information.

Cloud Privacy:

Cloud privacy refers to the right of individuals to control their personal information stored in the cloud, including knowing what data is collected about them, how it is stored, and how it is used or shared. It ensures that users have control over their information and protects it from misuse, unauthorized access, or disclosure.

In simple terms, cloud privacy is not only about keeping data confidential but also about giving users authority over their own data. If cloud privacy is not properly maintained, personal and sensitive information can be exposed or stolen through cyber-attacks and security breaches.

Cloud Compliance:

Cloud compliance refers to ensuring that cloud services follow all required laws, regulations, and standards set by organizations and governing bodies. It makes sure that data stored and processed in the cloud is handled securely and legally. Compliance in the cloud is a shared responsibility between the cloud provider and the customer, meaning both must follow proper rules and security practices.

Legal and Contractual Issues in Cloud Computing:

Legal and contractual issues refer to the problems or challenges that arise due to laws, regulations, and service agreements. These issues happen because cloud data is stored across different locations and managed by third-party providers, which can create confusion about responsibility, data protection, and legal control.

The legal and contractual issues are given as:



Data Ownership Confusion:

It is sometimes unclear who legally owns the data stored in the cloud—the user or the cloud provider. This can create disputes if data is misused or lost.



Data Privacy Laws

Different countries have different privacy laws. When data is stored across regions, it becomes difficult to follow all legal requirements properly.



Service Level Agreements (SLA) Problems:

An SLA is a contract between the user and the cloud provider. If the provider fails to meet agreed services like uptime, performance, or security, it may lead to conflicts or legal action.



Data Location Issues:

Cloud data is often stored in different countries. Some countries have strict rules about where data can be stored, which can create legal complications.



Liability Issues:

When a data breach or loss occurs, it may not be clear who is responsible—the cloud provider or the customer. This can lead to legal disputes.



Contract Termination Issues:

When a user stops using a cloud service, issues may arise about how data is safely returned, deleted, or migrated without loss or leakage.



Intellectual Property Rights (IPR) Issues:

Intellectual Property Rights (IPR) as a legal issue in cloud computing refers to the problems related to ownership, use, and protection of digital content such as software, data, documents, and applications stored in the cloud. Since cloud data is stored and shared through third-party providers, it can become unclear who legally owns the content and who has the right to use or modify it. This can lead to unauthorized copying, misuse of protected material, or copyright violations, creating legal disputes between users and cloud providers.

Risk Management in Cloud Computing:

Risk management in cloud computing is a process of identifying, analyzing, and controlling risks in cloud environments to prevent security threats and reduce their impact on data, applications, and services.

The various risk management techniques are:

1. Security monitoring
2. Incident Response Planning
3. Cloud Security Architecture design
4. Vulnerability Assessment
5. Data and Application Security
6. Virtual Machine Security

Security Monitoring:

Security monitoring is the continuous observation of cloud systems, networks, and applications to detect unusual or suspicious activities. It uses automated tools to identify threats in real time and helps prevent security incidents like data breaches or system attacks. It ensures the cloud environment remains safe and stable.

Incident Response Planning:

Incident response planning is a prepared plan that defines how an organization will detect, respond to, and recover from security incidents in cloud computing such as cyberattacks, data breaches, or system

failures. It is created before any incident happens so that the response is fast and well-organized when a problem occurs. The main goal is to reduce damage, restore services quickly, and protect important data.

Cloud Security Architecture Design:

Cloud security architecture design is the process of creating a structured plan to protect cloud systems, including data, applications, networks, and services. It defines how different security controls such as authentication, access control, encryption, and monitoring are implemented in the cloud environment.

This design ensures that security is built into the cloud system from the beginning, rather than added later. It also follows the shared responsibility model, where both the cloud provider and the user are responsible for maintaining security. The main goal is to create a safe, reliable, and well-protected cloud environment that prevents unauthorized access, data loss, and cyber threats.

Vulnerability Assessment:

Vulnerability assessment is the process of identifying weaknesses in cloud systems that could be exploited by attackers. It helps organizations understand security gaps and take corrective actions before attacks happen. This process improves overall cloud security and reduces risk.

Data and Application Security:

Data and application security focuses on protecting information and software in the cloud from unauthorized access, misuse, or attacks. It uses methods like authentication, encryption, access control, and security testing to ensure that both data and applications remain safe and reliable.

Virtual Machine Security:

Virtual machine security refers to protecting virtual machines (VMs) in cloud environments from cyber threats. It ensures that each VM is properly secured, monitored, and controlled. This includes applying security policies, protecting virtual networks, and preventing attacks in virtual and hybrid cloud systems.

Cloud Disaster Recovery:

[or, Cloud DR]

Disaster recovery in cloud computing refers to the use of cloud-based systems and services to back up, restore, and recover data, applications, and IT infrastructure after unexpected events such as cyberattacks, system failures, natural disasters, or data loss. In a cloud environment, critical data is continuously backed up and stored in secure locations so that it can be quickly restored when a disaster occurs. This helps organizations resume their normal operations with minimal disruption and ensures business continuity even during emergencies.

The benefits of Cloud Disaster Recovery are:

-  **Quick recovery:** Systems and data can be restored quickly after a failure.
-  **Reduced downtime:** Minimizes service interruptions and keeps business running.
-  **Data protection:** Ensures important data is safely backed up and not lost.
-  **Cost-effective:** Reduces the need for expensive physical backup infrastructure.
-  **Scalability:** Resources can be easily increased or decreased as needed.
-  **Business continuity:** Helps organizations continue operations during disasters.
-  **Remote accessibility:** Data can be recovered from anywhere through the cloud.

Disaster Recovery Planning in cloud computing:

Data recovery planning is the process of preparing a structured strategy to restore lost, corrupted, or inaccessible data from cloud systems. It ensures that important data is regularly backed up and can be recovered quickly in case of accidents, system failures, cyberattacks, or disasters. It ensures that important cloud data is regularly backed up and can be recovered quickly whenever needed. The main goal is to reduce data loss and maintain smooth business operations even during unexpected problems.

The best practices for Data recovery Planning are:

- ✚ Maintain regular and automated backups of important data.
- ✚ Store backups in multiple locations or regions for safety.
- ✚ Encrypt backup data to protect it from unauthorized access.
- ✚ Test recovery processes regularly to ensure they work properly.
- ✚ Define clear recovery time and recovery point objectives (RTO and RPO).
- ✚ Keep the recovery plan updated according to system changes.
- ✚ Use reliable cloud service providers with strong backup systems.

Disaster Management in cloud computing:

Disaster management is the process of identifying, responding to, and recovering from unexpected events such as cyberattacks, system failures, natural disasters, or data loss. It ensures cloud services & data are restored quickly and business operations continue with minimal disruption. It focuses on handling the situation during and after a disaster by using proper recovery plans and communication strategies.

The best practices for Disaster Management are:

- ✚ Maintain a well-documented and updated disaster recovery plan.
- ✚ Regularly test the disaster recovery plan to ensure it works effectively.
- ✚ Prioritize critical business applications and services during recovery.
- ✚ Ensure clear communication channels during emergencies.
- ✚ Focus on employee safety and coordination first during disasters.
- ✚ Use monitoring tools to detect disasters early.
- ✚ Coordinate with cloud service providers for fast recovery support.
- ✚ Continuously improve the disaster management plan based on past incidents.

Some Important Missing Topics

1. Explain Risk Assessment in Cloud.

Risk assessment in cloud computing is the process of identifying, analysing, and evaluating risks that may affect cloud systems, services, applications, and data. It helps organizations detect security threats, privacy issues, and operational problems before they cause damage. Risk assessment improves security, reduces data loss and cyberattacks, ensures business continuity, supports legal compliance, and creates a safer cloud environment.

The process of Risk Assessment in cloud computing is:

[Steps of Risk Assessment]

- 1. Identify Assets:** Determine important cloud resources such as data, applications, servers, and networks that need protection.
- 2. Identify Threats and Vulnerabilities:** Find possible threats and weaknesses like hacking, malware, weak passwords, or unauthorized access.
- 3. Analyze Risks:** Evaluate the possibility of threats occurring and the damage they may cause.
- 4. Evaluate Risk Level:** Classify risks as high, medium, or low based on their impact and probability.
- 5. Apply Risk Control Measures:** Use security methods like encryption, authentication, backups, and firewalls to reduce risks.
- 6. Monitor and Review Risks:** Continuously monitor cloud systems and regularly review risks because threats change over time.

2. Explain different building blocks of Data Centres.

Data centres are facilities used to store, manage, process, and distribute large amounts of data and applications. Different components work together to ensure smooth, secure, and reliable operation of the data centre.

The different building blocks of data centres are:

-  **Servers:** Servers are powerful computers that process data, run applications, and provide services to users and organizations.
-  **Storage Systems:** Storage devices are used to store data, files, databases, and backups securely for easy access and recovery.
-  **Networking Components:** Components such as routers, switches, and cables connect servers and devices to enable communication and data transfer.
-  **Power Supply System:** Includes UPS, generators, and power distribution units that provide continuous electricity and prevent downtime during power failures.
-  **Cooling System:** Cooling and air conditioning systems control temperature and humidity to prevent overheating of servers and equipment.
-  **Security System:** Physical and digital security measures such as CCTV cameras, firewalls, biometric access, and encryption protect the data centre from unauthorized access and attacks.
-  **Backup and Disaster Recovery System:** Backup systems create copies of data and disaster recovery systems help restore services quickly during failures or disasters.
-  **Management and Monitoring Tools:** These tools monitor performance, network traffic, power usage, and system health to ensure efficient operation of the data centre.

3. What is host security? Explain different approaches for host security.

Host security refers to protecting individual computers or servers (hosts) in a network or cloud environment from unauthorized access, malware, attacks, and misuse. It ensures the safety of the operating system, applications, and data running on a host so that the system remains secure and reliable.

Host security is important because a single compromised host can affect the entire network or cloud system. It improves overall security, protects sensitive data, reduces cyberattacks, ensures system stability, and supports continuous and reliable service availability.

The different approaches of Host security are:

- 🚦 **Operating System Security:** Keeping the OS updated with security patches, configuring secure settings, and disabling unnecessary services to reduce vulnerabilities.
- 🚦 **Authentication and Access Control:** Using strong passwords, multi-factor authentication, and role-based access control (RBAC) to ensure only authorized users can access the host.
- 🚦 **Antivirus and Anti-malware Protection:** Installing security software to detect, block, and remove viruses, malware, and other harmful programs.
- 🚦 **Firewall Protection:** Using host-based firewalls to monitor and control incoming and outgoing network traffic based on security rules.
- 🚦 **Encryption:** Protecting sensitive data by converting it into unreadable form so that unauthorized users cannot access it even if they get the data.
- 🚦 **Patch Management:** Regularly updating software, applications, and the operating system to fix security bugs and vulnerabilities.
- 🚦 **Logging and Monitoring:** Tracking system activities and user actions to detect suspicious behavior and respond to security incidents quickly.
- 🚦 **Backup and Recovery:** Regularly backing up important data and having recovery plans to restore the system after attacks or failures.

4. Explain single-purpose architecture to multi-purpose architectures.

Single-purpose architecture is a system designed to perform only one specific task or service. Each server or system is dedicated to a single function, such as web hosting, database management, or file storage. It is simple, easy to manage, and provides good performance for that specific task, but it is less flexible and not efficient when workload changes or multiple services are needed.

Multi-purpose architecture is a system where a single infrastructure can handle multiple tasks or services at the same time. A single server or cloud system can run different applications like web services, databases, storage, and analytics together. It is more flexible, scalable, and cost-efficient, but it is more complex to manage compared to single-purpose systems.

5. What are different factors involved in disaster recovery planning?

Disaster recovery planning (DRP) is the process of preparing strategies to restore IT systems, data, and operations after a disaster such as system failure, cyberattack, or natural calamity. A good DR plan ensures quick recovery with minimum data loss and downtime.

The factors involved in Recovery Planning are:

1. **Risk Assessment:** Identifies possible threats like hardware failure, cyberattacks, power outages, or natural disasters that can disrupt systems.

- 2. Business Impact Analysis (BIA):** Determines critical systems and evaluates the impact of their failure on business operations.
- 3. Recovery Objectives (RTO & RPO):** Defines Recovery Time Objective (maximum acceptable downtime) and Recovery Point Objective (acceptable data loss limit).
- 4. Backup Strategy:** Plans regular data backups and decides storage locations such as cloud or off-site systems.
- 5. Disaster Recovery Site:** Selects backup environments like hot, warm, or cold sites to restore operations during emergencies.
- 6. Data Security Measures:** Ensures backups and recovery systems are protected using encryption and access controls.
- 7. Communication Plan:** Establishes clear communication channels to inform users, employees, and stakeholders during a disaster.
- 8. Testing and Maintenance:** Regularly tests and updates the DR plan to ensure it works effectively when needed.

6. Mention some popularly used cloud computing services.

Cloud computing services are different types of services delivered over the internet.

The services are:

a. Software as a Service (SaaS):

Ready-to-use software accessed through the internet without installation.

Example: Gmail, Google Drive, Google Docs, Canva, Netflix, etc.

b. Platform as a Service (PaaS):

Provides a platform to develop, test, and deploy applications without managing hardware.

Example: Google App Engine, Microsoft Azure App Service, AWS Elastic Beanstalk, Firebase.

c. Infrastructure as a Service (IaaS):

Provides virtual machines, storage, and networking resources over the internet.

Example: Amazon EC2, Google Compute Engine, Microsoft Azure Virtual Machines, etc.

d. Communication as a Service (CaaS):

Provides real-time communication services like voice, video, messaging over cloud.

Example: Zoom, Microsoft Teams, Google Meet, Skype, etc.

e. Monitoring as a Service (MaaS):

Provides cloud-based monitoring of systems, networks, and applications.

Example: AWS CloudWatch, Google Cloud Monitoring, etc.

7. Security issues which should be discussed with cloud-computing vendor?

When using cloud services, it is important to clearly discuss security issues with the cloud vendor to ensure data safety, privacy, and reliable service. We must discuss following things.

-  **Data Security:** Ask how your data is protected using encryption (both in storage and transmission) and who can access it.
-  **Access Control:** Ensure the vendor provides strong authentication methods like passwords, multi-factor authentication, and role-based access control.
-  **Data Privacy:** Clarify how your data is stored, processed, and whether it is shared with any third party.

- 🔗 **Data Backup and Recovery:** Confirm how often backups are taken and how quickly data can be restored after failure or attack.
- 🔗 **Compliance and Legal Issues:** Check whether the vendor follows security standards and laws like ISO, GDPR, or other regulations.
- 🔗 **Incident Response:** Understand how the vendor responds to security breaches or cyberattacks and how quickly they inform users.
- 🔗 **Service Availability (Uptime):** Discuss guarantees for uptime and protection against downtime or service disruptions.
- 🔗 **Physical Security of Data Centers:** Ensure data centers are protected from unauthorized physical access, theft, fire, or natural disasters.

8. Differences between:

Traditional Corporate Data Centre	Cloud Data Centre
➤ Owned and managed by a single organization on its own premises.	➤ Owned and managed by cloud service providers and accessed over the internet.
➤ High initial cost for hardware, setup, and maintenance.	➤ Pay-as-you-go model with low initial cost.
➤ Limited scalability; expanding resources takes time and cost.	➤ Highly scalable; resources can be increased or decreased instantly.
➤ Requires in-house IT staff for maintenance and management.	➤ Managed by cloud provider with minimal user-side maintenance.
➤ Resources are fixed and may remain underutilized.	➤ Resources are flexible and efficiently shared among users.
➤ Recovery from disasters is slower and complex.	➤ Faster disaster recovery with built-in backup and redundancy.
➤ Accessibility is usually limited to internal network or VPN.	➤ Accessible from anywhere via the internet.
➤ Upgrading hardware is costly and time-consuming.	➤ Automatic and continuous upgrades by provider.

9. Explain disaster recovery process in cloud computing.

Disaster recovery process is the actual set of actions taken to restore cloud systems, data, and services after a failure or disaster occurs. It focuses on quickly bringing services back online with minimum downtime and data loss.

The disaster recover process is given as:

1. **Disaster detection:** Identify system failure such as server crash, cyberattack, network failure, or hardware breakdown.
2. **Alert and notification:** Inform the IT team and concerned authorities about the system failure or service disruption.
3. **Failover activation:** Switch operations from the primary system to a backup or disaster recovery site to keep services running.

4. **Data recovery:** Restore lost or damaged data using backup copies stored in cloud storage.
5. **System restoration:** Restart applications, databases, and services in the backup environment. Backup environment refers to a secondary setup that stores copies of systems and data to take over operations during failure.
6. **Verification and testing:** Check that all services are working correctly and ensure data accuracy and system performance.
7. **Failback (Return to primary system):** Shift operations back to the original system after it is repaired and stable.

10. Difference between parallel computing and distributed computing?

Parallel Computing	Distributed Computing
➤ Multiple processors work together on a single problem at the same time.	➤ Multiple computers work together over a network to complete a task.
➤ All processors are usually in the same system or closely connected.	➤ Computers are located in different places and connected through a network.
➤ Uses shared memory in most cases.	➤ Each computer has its own separate memory.
➤ Mainly focuses on increasing processing speed.	➤ Mainly focuses on sharing workload and improving scalability.
➤ Communication between processors is very fast.	➤ Communication is slower because it depends on network.
➤ Best suited for scientific and high-performance computing tasks.	➤ Used in cloud computing, web services, and large-scale systems.
➤ Example: Supercomputers, multi-core processors.	➤ Example: Cloud platforms, distributed databases, Google services.

11. How does the monitoring agent monitor cloud usage?

A monitoring agent is a software tool installed on cloud systems such as servers, virtual machines, or applications that continuously collects and reports data on resource usage and system performance. It helps monitor how cloud resources like CPU, memory, storage, and network are being used, enabling providers and users to detect issues early and ensure efficient system performance.

The Monitoring agent monitors cloud usage by:

-  **Data collection:** Collects resource usage data such as CPU usage, memory usage, storage usage, network traffic, and application performance.
-  **Continuous monitoring:** Continuously tracks cloud resources at regular intervals to observe system behavior in real time.
-  **Performance tracking:** Measures how efficiently servers and applications are working under different workloads.
-  **Usage reporting:** Sends collected data to a central monitoring dashboard or system for analysis.
-  **Alert generation:** Generates alerts when resource usage is too high, too low, or when any abnormal activity or failure is detected.

- 🚦 **Resource optimization support:** Helps identify underused or overloaded resources so they can be adjusted for better performance and efficiency.

12. How does cloud-native foundation define cloud-native applications?

Cloud-native applications are applications designed to fully use cloud computing features such as scalability, flexibility, and resilience. According to the Cloud Native Computing Foundation (CNCF), cloud-native applications are built using microservices architecture, run in containers, are dynamically managed through orchestration systems, and are designed to be easily scalable, portable, and continuously updated.

In simple terms, cloud-native applications are modern applications built specifically for the cloud environment so they can run efficiently, scale quickly, and recover easily from failures.

13. How resource replication takes place in cloud computing?

Resource replication in cloud computing is the process of creating and maintaining multiple copies of data, applications, or resources across different servers or data centers to ensure high availability, reliability, and fault tolerance. It works by copying data from a primary system to one or more secondary systems either in real time (synchronous replication) or with a slight delay (asynchronous replication).

These replicas are often stored in different geographic locations to protect against failures or disasters. If the primary resource fails, the system automatically switches to a replica through a failover mechanism, ensuring continuous service and minimal downtime.

14. Difference between scalability and elasticity?

Scalability	Elasticity
➤ Scalability is the ability of a system to increase or decrease resources to handle growing workload.	➤ Elasticity is the ability of a system to automatically adjust resources based on current demand.
➤ Changes are usually planned and long-term.	➤ Changes are automatic and real-time.
➤ Focuses on handling future growth.	➤ Focuses on handling sudden or temporary changes in demand.
➤ Resource adjustment is slower.	➤ Resource adjustment is fast and dynamic.
➤ Mostly managed manually or semi-automatically.	➤ Fully automated in cloud systems.
➤ Example: Adding more servers to handle increasing users over time.	➤ Example: Automatically increasing servers during traffic spikes and reducing them when traffic drops.

15. What are the implementation issues on SaaS? Explain.

SaaS implementation refers to deploying and using software applications over the internet. Although it is easy to use, there are several challenges that organizations may face during implementation.

- 🚦 **Data security and privacy issues:** Data is stored on the provider's servers, so there is a risk of unauthorized access, data leakage, or privacy concerns.
- 🚦 **Internet dependency:** SaaS applications require a stable internet connection; without it, services cannot be accessed.

- ✚ **Integration problems:** Difficult to connect SaaS applications with existing legacy systems or other software used by the organization.
- ✚ **Limited control:** Users have less control over software updates, infrastructure, and system configuration.
- ✚ **Performance issues:** Speed and performance may vary depending on internet quality and provider server load.
- ✚ **Compliance challenges:** Meeting legal and regulatory requirements can be difficult when data is stored in different regions.
- ✚ **Limited customization:** SaaS applications may not fully meet specific business needs due to fixed features.

16. Describe possible services that can be achieved through IaaS.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet such as servers, storage, and networking. It allows users to rent and manage IT infrastructure without buying physical hardware, making systems flexible and cost-effective.

The services that can be achieved through IaaS are:

- ✚ **Virtual machines (compute services):** Provides virtual servers where users can install operating systems and run applications as needed.
- ✚ **Storage services:** Offers scalable storage for data, backups, and files.
- ✚ **Networking services:** Provides virtual networks, IPs, and secure connectivity between resources.
- ✚ **Backup and recovery services:** Enables data backup and restoration during failures.
- ✚ **Load balancing services:** Distributes traffic across multiple servers for better performance.
- ✚ **Security services:** Includes firewalls, encryption, and access control for protection.
- ✚ **Scalability services:** Allows easy increase or decrease of resources based on demand.

17. What do you mean by Elastic IP Addressing?

Elastic IP Addressing refers to a static public IP address in cloud computing that can be dynamically reassigned to different virtual machines or instances. It is called “elastic” because it can be quickly attached or detached from resources based on need, without changing the IP address itself.

18. Describe How Elastic IPs work in cloud services.

Elastic IP works by first assigning a static public IP address to a user account in the cloud. This IP is then attached to a running virtual machine (instance) so it can be accessed over the internet. If the instance stops, fails, or is replaced, the same Elastic IP can be quickly detached and reassigned to another instance. This allows services to remain reachable without changing the IP address, ensuring continuous access and flexibility.

19. Explain Network Intrusion Detection.

Network Intrusion Detection is a security process used to monitor network traffic and identify any unauthorized or suspicious activities. It helps detect attacks such as hacking attempts, malware, unauthorized access, or abnormal data transfers in a network or cloud environment.

It works by continuously analyzing incoming and outgoing network packets and comparing them with known attack patterns (signature-based detection) or normal behaviour patterns (anomaly-based detection). If any unusual or malicious activity is found, the system generates an alert to inform administrators so that they can take action immediately.

Network Intrusion Detection improves security by detecting threats early, preventing data breaches, and ensuring safe and reliable network communication.

20. Explain the concept of implementing network intrusion detection in the cloud.

Network Intrusion Detection in the cloud is implemented by continuously monitoring and analysing network traffic within cloud environments to detect suspicious or unauthorized activities. It is done by deploying Intrusion Detection Systems (IDS) as virtual security tools on cloud networks, virtual machines, or at the network gateway.

In implementation, cloud IDS collects network packets, logs, and traffic data from different cloud resources. It then analyses this data using signature-based detection (matching known attack patterns) and anomaly-based detection (finding unusual behaviour compared to normal traffic). If any threat such as hacking attempts, malware, or abnormal traffic is detected, the system generates real-time alerts and sends them to administrators or security tools for quick response.

Cloud-based IDS can be centrally managed and scaled easily, making it suitable for large and dynamic cloud environments. It helps improve security, detect attacks early, and protect cloud data and applications from unauthorized access.
